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A development of the Trangia Container Supports

A redesign of the container supports used in the Trangia Storm Kitchen

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Abstract

This project was conducted in conjunction with Trangia, through the Royal Institute of Technology in Stockholm. Trangia is a Swedish company producing camping equipment, mainly known for "Stormköket". The purpose of the project was to redevelop the container supports for the Trangia Storm kitchen.

The project began with a literature study. This included understanding relevant regulations for outdoor stoves. It also included studying feasible materials for the product use case. The next stage in research was understanding Trangia's position in the existing market and its customer base. This was found through product evaluation of their design expression and brand guidelines.

This was followed by several rounds of experiments where the optimal height for heat transfer of the cooking containers was investigated. This resulted in a lowered height for the pan, but the optimal pot height led to higher amounts of soot.

Through broad and then converging ideation through sketching, a design was developed. This led to 3D printed and subsequently metal prototypes. Through a vast number of iterations the design was refined into the 180 flag concept. This design has been developed to the completion for manufacture and assembly to a theoretical level. Further refinement is necessary to implement the product, but the functionality has been proven through several prototypes.

Keywords:

Trangia, Storm Kitchen, Camping stove, Camping, Outdoor cooking

Sammanfattning

Detta projekt genomfördes i samarbete med Trangia, genom Kungliga Tekniska Högskolan i Stockholm. Trangia är ett svenskt företag som tillverkar campingutrustning, främst känt för "Stormköket". Syftet med projektet var att förbättra kittelstöden till Trangias Stormkök.

Projektet inleddes med en omfattande litteraturstudie. Detta inkluderade en undersökning av relevanta regler för campingkök, samt lämpliga material för produkten. Nästa steg i studien var att förstå Trangias position på den befintliga marknaden, och deras kundbas. Detta undersöktes genom en utvärdering av produkternas designuttryck och Trangias varumärkesriktlinjer.

För att förstå produkten ytterligare inleddes flera experimentomgångar där den optimala höjden för värmeöverföring till kokkärlen undersöktes. Detta resulterade i en sänkt höjd för stekpannan, men den optimala kastrullhöjden ledde till högre mängder sot än önskat.

Genom bred och sedan konvergerande idégenerering med hjälp av skisser utvecklades en design. Prototyper utvecklades sedan med hjälp av 3D-skrivare, som därefter förfinades genom metallprototyper. Ett stort antal iterationer optimerade designen vilket slutligen resulterade i 180-flaggkonceptet. Produkten har utvecklats fram till tillverkning och montering på en teoretisk nivå. Ytterligare utveckling är nödvändig för att implementera produkten, men funktionaliteten har bekräftats genom flera prototyper.

Nyckelord:

Trangia, Stormkök, Campingkök, Camping, Utomhusmatlagning

Thursday 19th June, 2025

Acknowledgements

Thank you Conrad! Your fun shirts have kept us going these past 5 months.

Thanks also to Helena for supporting us throughout the project. Your assistance and provided resources has helped us create a thorough and developed design.

Stockholm, June, 2025 Victor Sandberg and Theo Sloos

Nomenclature

- CAD - Computer Aided Design
- CS - Container Supports. Used to hold cooking containers in the Trangia storm kitchen.
- DofE - Duke of Edinburgh award
- DFA - Design for Assembly
- DFM - Design for manufacturing
- EU - European Union
- FEA - Finite Element Analysis
- LWS - Lower Wind Shield
- mm - Millimetres
- Primed/priming - stage between the spirit burner being lit and flame emitting from the radial holes
- Sparring - Trangia made simmer ring
- TSK - Trangia Storm Kitchen
- Tamoj - Trangia made handle for picking up pots and pans
- UWS - Upper Wind Shield
- Waterjet cutting- Cutting metal with a high pressure jet of water.

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1 Introduction

The following section will cover the background of the company, the addressed issues, and limitations of the given assignment.

1.1 Background

Trangia is a 100 year old company based in the north of Sweden, providing outdoor cooking equipment. The designs are aimed to be compact and usable for long distance outdoor trekking and camping. The company largely focuses on durability and the implication that their products can be handed down through generations [1]. Their traditional storm kitchen, first introduced in the 1950's, uses stacking to contain all of the parts for transport. See Figure 1 below for an unpacked storm kitchen.



Figure 1: *The Trangia storm kitchen with spirit burner*

1.2 The Brief

In collaboration with Trangia, a proposed redesign of the storm kitchen's CS shall be presented.

To hold a kettle or pan, the storm kitchen uses three CS which can fold out. See Figure 2 for the current container supports.



Figure 2: *The current container supports of the TSK*

These supports need to be simplified, as they require several instances of fine tolerances for manufacture. The aim is to produce an alternative which requires less complex manufacturing, but the proposal can drastically differ from the current design. It does however, still need to be compatible with all previous versions of the TSK.

Trangia provide two options for heat source on the TSK. The traditional standard option is the Spirit burner, which has a Sparring for simmering water and a lid with a gasket. The gas burner is fixed into the LWS at its aperture. Both options can be seen in Figure 3 below.



(a) *The Trangia spirit burner in the middle with Sparringen to the left and burner lid to the right*



(b) *The Trangia Gas burner*

Figure 3: The two options of heat source for the TSK

1.3 Limitations

The overall design of the Trangia storm kitchen has remained the same since 1951. An overarching limitation during this project was therefore that all new solutions had to be compatible with previous versions of the storm kitchen.

2 Literature Study

Within this section, past literature is established to understand the current academic scope as well as industrial manufacturing knowledge. Additionally relevant regulations and laws are considered as a basis of understanding of health and safety requirements and sustainability. Though the CS wont be in direct contact with food, it could be affected from heat and create fumes.

2.1 EU regulations for food safe materials

Currently, the EU has regulation from 2004, stating safe practices that can be considered no longer appropriate. In the future, the EU intends to create regulations to assist in designs developed for use in contact with food. This is to assist in the development of new innovation in the field which likely can help in promoting future sustainable practices for safe reusable and recyclable packaging [2].

Currently, within the regulations and amendments in the last year, thousands of synthetic chemicals have been banned as they are considered to be environmental pollutants. Many of these chemicals have been found to have adverse effects on humans as well [2]. As there is a vast number of dangerous chemicals it will be more appropriate to consider proposed chemicals against regulations as each case occurs during the project.

Within the 2004 regulation from the EU, it states that the rules cover a range of materials from Adhesives, metals and alloys to wood. The overall consideration of the regulation is to ensure no endangering of human health, change to the foods composition or have a sensory effect (organoleptic) to the food [3].

A potential source for finding authorised materials and chemicals can be found on the ECHA website. Listed is over one thousand plastics used for food contact [3].

2.2 EU regulations on sustainability

The Trangia Storm Kitchen can be considered a highly sustainable product, where the design has been rarely changed since its inception. This has guaranteed users the convenience to fix and replace parts as needed, increasing the longevity of their TSK. The product also uses recyclable materials in its design using aluminium and steel for the main components [4]. With this in mind it is highly important to consider how any changed parts or materials in the TSK could impact its Sustainable pedigree.

The EU has created a directive helping companies and designers with 'establishing a framework for the setting of Eco-design requirements for sustainable products'. This legislation highlights the use and need for a circular economy, aim for climate neutrality and sustainable consumption, amongst others [5].

As with the previous literature area, the regulations are considerably large for sustainable design practices. It can be considered within the scope of the project but likely driven through supervision from Trangia.

2.3 Predominant manufacturing materials analysis

In order to better understand the decision making for producing the TSK, the materials used along with other common production materials were evaluated. This included: Tensile strength, Vickers hardness, melting point, price (by weight and volume), thermal conductivity and Density. These were considered to be the most relevant properties for the product use case. The results in the Table 1 below were found from the ANSYS source: Granta Edupack.

Table 1: Properties of common production materials

CPM's	**Tensile Strength MPa**	**Hardness HV**	**Melting point °C**	**Cost SEK/Kg**	**Cost SEK/m3**	**Thermal conductivity W/m°C**	**Density kg/m3**
Steel 1020	395	120	1500	14	0.13000	52	7850
Aluminium 5005 H14	106	40	660	35.7	96000	201	2690
Titanium (grade 11)	298	120	1655	210	963000	20	4510
Cast Iron austenite	190	178	1200	51.4	379000	46	7300
Brass (yellow)	333	70	850	80	677000	121	8400
Tin (copper)	44.3	9	264	354	2500000	63.1	7320

Through comparison of the materials, Trangia's material choices have become clear. The use of aluminium and steel is economically wise, plus their properties assist for the use case. Aluminium is highly conductive, which is helpful for heating the pots and pans. Additionally, Steels high tensile strength provides durability, which ensures that Trangia's brand marker of longevity continues. It is understandable that Trangia is not using Titanium, as the price nullifies all advantages.

2.3.1 Degradation of aluminium and steel in outdoor environments

With regards to Trangia, longevity is the backbone of their branding and key cause for why customers use their products. Understanding how this longevity is ensured

for the materials is key, especially when deciding how to design a replacement for the CS.

Within Myer Kutz's,'Handbook of Environmental Degradation[6] , information can be found on how high heat can impact materials. This is highly relevant in the context of the TSK. While cooking it can be important to consider degradation of the aluminium. For materials containing iron, the potential for electromagnetic enhanced corrosion is higher in cooking, as Kutz explains the 'corrosion rate of low carbon steels is very fast with hydrochloric acid'. As on Trangia website it states the importance to limit use of 'highly acidic or salty foods' , this can remove the aluminium onto the food [4] . The Hard anodized aluminium is intended to solve such issues, plus adding strength.

For the supports the most likely materials used will be stainless steel or aluminium as they are commonplace in Trangia components, Stainless steel is however primarily used as part of their proprietary material: Duossal.

3 Background Research

To better understand the TSK, further background researched was required. This has pertained primarily to the testing and analysing of the TSK's components. Additionally the highlighting of competitor products and the existing market is presented within this section.

3.1 Existing Storm Kitchen components

The standard storm kitchen is made up of eight parts. Every set comes with a frying pan, along with two sizes of pots that can be stacked together. The wind shield is comprised of an upper and a lower part. To lift the pots and pan a handle, called a Tamoj, is included. One can choose the type of burner that one prefers, and in its collapsed position the kitchen is held together by a strap. The main parts are shown in Figure 4 below.

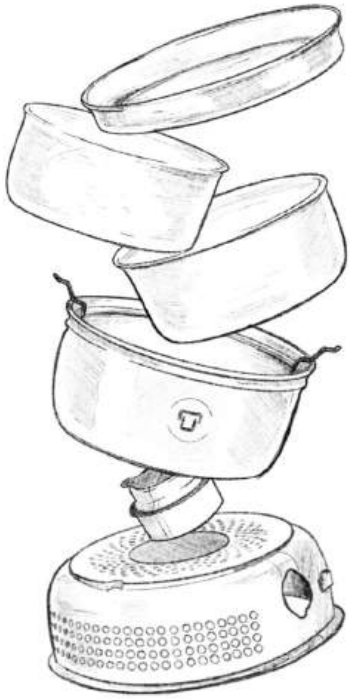


Figure 4: *Main parts of the storm kitchen. From the top: frying pan, small pot, large pot, upper wind shield, spirit burner, lower wind shield*

3.1.1 Component weight

The Trangia Storm kitchen is considered to be a robust product, it however is considered to be quite heavy. To better understand how the mass of the product is distributed, the relevant components were measured. This can be used to try and ensure any changes of the device in weight is minimal. Listed in Table 2 below are the relevant components and their according weight.

Table 2: Component Weights

Number	Component	Weight (g)
1	Ring 25	59.7
2	Ring 27	49.9
3	CS	6.7
4	Sparring	22.7
5	Burner lid + o ring	21.0
6	Spirit burner	66.8
7	Combined Ring 25	79.8
8	Combined Ring 27	70.0
9	Combined burner	110.5

3.2 Testing of equipment

The testing of the equipment was performed in several environments and situations, to better understand the user requirements from a first hand perspective.

The initial testing of the equipment was performed in a garden as well as on a short ice skating trip. In Figure 5 below testing of the TSK is shown in the garden.

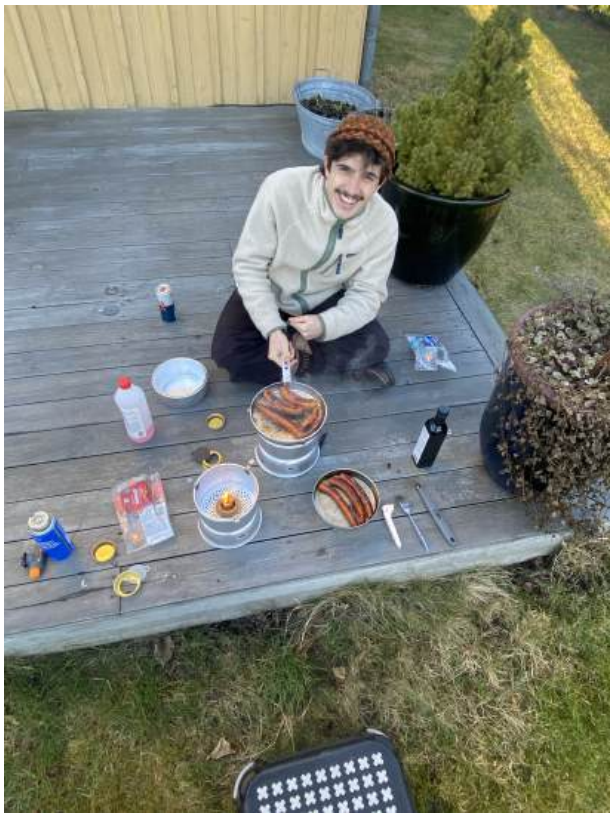


Figure 5: *Testing the Trangia Storm kitchen in a garden (convenient environment)*

In the following, Figure 6 below, a second trial of the TSK was done. Sausages were again cooked, but on a long distance skating trip. The simmer ring was used, but had to be repeatedly removed then replaced throughout the process. This made holding a good level of flame difficult and tiresome. The TSK was convenient for the job however, as it easily fit within a rucksack packed for skating. An additional observation was the way the TSK behaved when packed. Since the pots, burner and Tamoj are loose inside the wind shield when stored, they make a noise when moving. This became apparent when skating with the TSK in a rucksack, and the noise was an unwelcome distraction.



Figure 6: *Cooking on the edge of a lake (inconvenient environment)*

Listed are the main takeaways from the testing process:

- **1**: It is very difficult to find the perfect level of heat. An open flame is too high and with the simmer ring it is too low.
- **2**: It is difficult to adjust the simmer ring, having to remove it with a fork and then wait for it to cool.
- **3**: The Tamoj is excellent.
- **4**: While travelling (walking, skating) the Trangia Storm kitchen can rattle.
- **5**: It takes far longer to prime the burner in the cold.

- **6**: If the lid of the burner is screwed on before it fully cools down, the o-ring can melt.
- **7**:Even though the sparring is brass and so, thermally conductive, it still takes a while to cool, and shows no sign for when it is safe to touch.
- **8**: The CS functioned quite well, they could however, have fallen down. This would especially be the case on a non flat surface.

3.2.1 Dimension limitations

To focus the ideation stage and limit the scope, dimensions of the Trangia were taken. The dimensions measured which relate to the CS are shown in Figure 7 below.

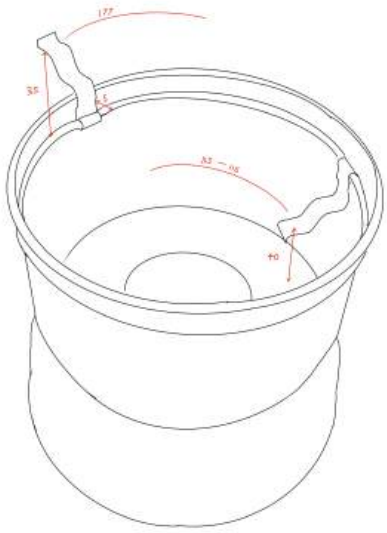


Figure 7: *Dimension requirements of the Trangia Storm, all units are in millimetres*

It is noted, that these dimensions relate primarily to the relationship between the containers and the CS. The container support itself isn't dimensioned as it is expected that the dimensions will not be relevant outside its relationship to other components.

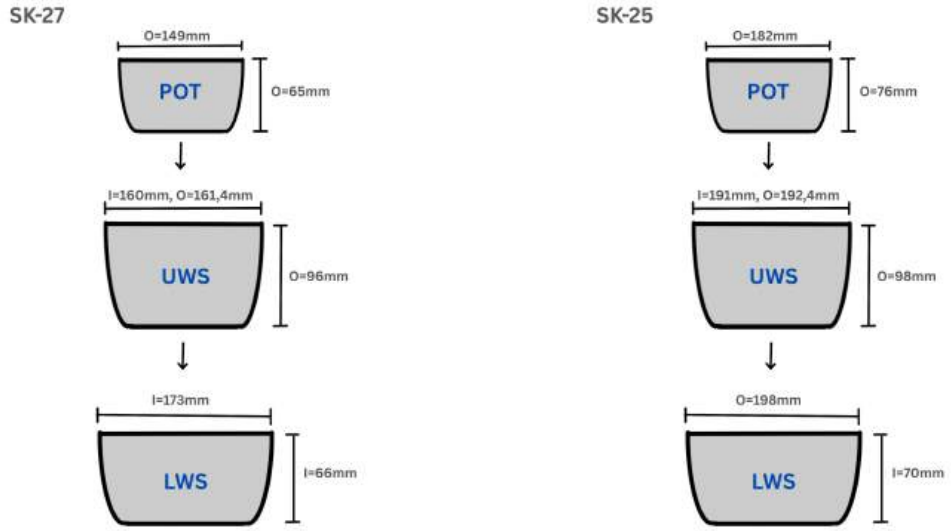
4 Design limitations

To focus the ideation phase, the specification was enhanced. Through this chapter, the function guidelines and specific physical requirements have been stated.

4.1 Geometrical limitations

During the Trangia factory visit, the scope was further refined but also broadened through discussions on how the CS could be changed. It has been stated that the new design can drastically alter the UWS. This is provided that the design continues to be compatible with previous versions. However, the remaining components are to stay the same.

In Figure 8 below are the dimensions and relationship between the current UWS and other components for Storm kitchen 27 and 25. These dimensions are the geometrical limitations that any future CS design must relate to.



(a) *27 TSK dimensions* (b) *25 TSK dimensions*

Figure 8: 25 and 27 Storm kitchen dimensions

The required dimensions for any design proposal can be calculated by finding the delta between the components. In the equation below are the deltas between the LWS and the UWS, and the UWS and the Pot. The UWS must have its outer (O) and inner (I) values distinguished, as the two values relate to the LWS and Pot respectively. The difference in values is based off the component's wall thickness, which is 0.7mm.

The height's Delta formulas:

$$\text{LWS height} - \text{UWS height} = \Delta_1 \tag{1}$$

$$\text{UWS height} - \text{POT height} = \Delta_2 \tag{2}$$

The result of these height variation formulas can be seen in table 3.

Table 3: Height variation of UWS,LWS and Pot

Δ	TSK size	variation (mm)
1	27 25	30 28
2	27 25	31 22

The diameter's delta formulas:

$$\text{LWS Diameter} - \text{UWS O Diameter} = \delta_1 \tag{3}$$

$$\text{UWS I diameter} - \text{POT diameter} = \delta_2 \tag{4}$$

The result of these diameter variation formulas can be seen in table 4.

Table 4: Diameter variation of UWS,LWS and Pot

δ	TSK size	variation (mm)
1	27 25	11,6 5,6
2	27 25	11 9

The values found in the tables above are from measurements made with a ruler. They can be considered accurate to within 0.25 millimetres, with the expected maximum inaccuracy to be 0.5 millimetres, as two of these values combined form a Delta value. This is considered allowable as the values are more to the benefit for general design ideas, not manufacture.

These dimensions are based on the two functions of the UWS. The UWS, firstly, houses the two pots. Any new UWS must house the two pots and then additionally the Tamoj and heating implement. During this function the UWS must fit within the LWS, the added height above the top of the LWS could also be a factor for future designs.

The second function of the UWS is during cooking. The UWS is used to attach the CS, and therefore the pots and pans. This means the UWS must facilitate the two heights, through the CS or any new proposal.

When building the Storm kitchen the LWS and UWS are connected by interlocking the two parts. For any proposed ideas, this feature must be kept to ensure backwards compatibility.

A final design limitation is that the UWS must still protect against the wind. It is then expected that if the new UWS has holes, they will not inhibit this function.

For a successful design proposal, it must function for both the 25 and 27 TSK.

4.2 Manufacturing limitations

For this design development, the success of the proposed change can be measured by how quick the manufacturing process is. The current process of implementing the ring and stamped CS takes a certain number of disclosed seconds per TSK.

For every second that is reduced to the process, Trangia can save a certain amount over thousands of Storm kitchens each year. The more seconds that the new design saves, the larger the budget for investment.

The budget and calculations have been presented but are withheld in this report for company privacy.

4.2.1 Collation of design limitations

Listed below are the design limitations previously stated within this chapter:

- The new solution must allow the UWS to pack within the LWS, keeping similar deltas to the current solution.
- The new solution must allow the UWS to house the Pots, Tamoj and gas burner/spirit burner, keeping similar deltas to the current solution.
- The new solution must allow support for cooking of the pan and pots.
- The silhouette of the new solution should be congruent with the Trangia brand and form.
- The new solution must be faster to manufacture compared to the current solution.

These limitations along with the specification will direct the Implementation and ideation stage.

4.3 Height heat exchange experiments

It was deemed that there could be potential to test how the height of the pot from the flame, could affect the effectiveness of the flame. This information is of interest as

with the CS, a future design could be more effective with a different height from the flame. After discussions with Trangia it became apparent that the current position had not been reviewed previously.

4.3.1 Height experiment 1

For this experiment, the Trangia 27 with its smallest pot was used. For the flame, a bottle of spirit fuel was bought from Naturkompaniet, an image of this fuel can be seen in figure 9 below.



Figure 9: *Online image of the spirit fuel used for the experiments*

To prepare for the experiment, 3 millimetre thick cardboard was cut into rings. Large rings first increase the distance from the flame origin, layering them beneath the UWS. After three layers added, they are then removed and replaced up to three layers underneath the simmer ring. This decreases the distance from the pot to the flame. The normal distance is also measured as the control.

Below is listed the method run for the experiment:

- **Step 1**: Place cardboard underneath the UWS
- **Step 2**: Fill small pot with 200ml of cold water
- **Step 3**: Measure initial temperature
- **Step 4**: Place above the open flame on the container supports
- **Step 5**: Measure temperature after 120 seconds
- **Step 6**: Measure time until boiling
- **Step 7**: Repeat the previous steps adding a cardboard insert under the UWS.
- **Step 8**: Removing the cardboard from under the UWS. Repeat steps 1 through 6 then adding cardboard inserts under the spirit burner.

In Table 5 the results of the experiment are shown. Unfortunately during the experiment it became apparent that the flame of the spirit burner was burning the

cardboard in both instances when there were three layers of cardboard. The issue is shown in Figure 10 below.

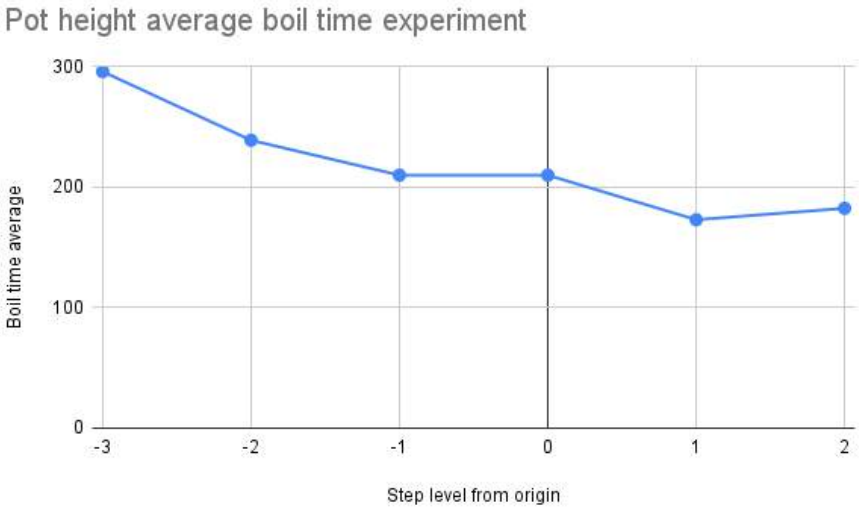
Table 5: Height heat exchange experiment

Step distance from origin	**start temp avg, °C**	**120 second temp avg, °C**	**Time to Boil, s**
2	20.55	77.35	182
1	19.10	76.97	173
0	19.07	60.30	210
-1	19.87	74.07	210
-2	19.87	57.93	239
-3	20.6	51.5	296

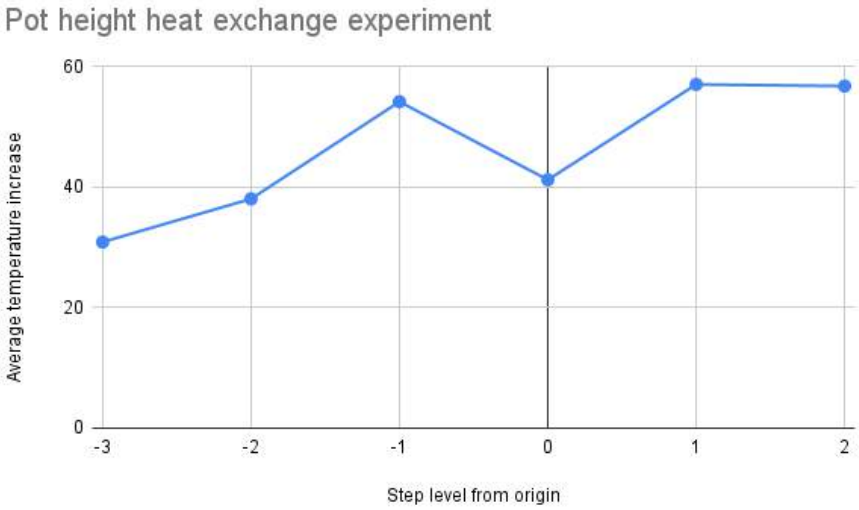


Figure 10: *Third layer of cardboard, unfortunately on fire*

The table above only shows the average values, the complete data is within Appendix C. As previously mentioned, the results are not conclusive, as for step -3 from origin, only one result was taken. In Figure 11a and 11b below the results are presented within graphs.



(a) *Boil time*



(b) *Temperature after 120 seconds*

Figure 11: Boil time vs Pot height and heat exchange over 120 seconds vs Pot height (3 millimetre increments)

In Figure 5 , there is a clear trend where the further the pot is from the flame the faster the water came to boil. In Figure 11b the trend is not quite as clear. Though it does appear that a greater distance increases the temperature after 120 seconds, -1 appears to be an anomaly. We did the origin/control test first and then -1. The first test, perhaps, increased the potential flame for the -1 step test.

An additional limitation for the test was that the flame could not be consistent. The fire had to be put out to safely add the layers of cardboard. This likely reduced the validity of the experiment.

The cardboard spacers covered some of the air holes during testing. It was observed that the flame behaved differently with the spacers. It was, however, difficult to conclude if this was due to the burner increasing in temperature or if the air flow had been altered.

This experiment does appear to show a correlation between an increased height and speed at heating water. The time difference however is not especially large. Plus considering the CS is used for two heights, there is potential that a shorter height will benefit the pot but hinder a pan when the CS is flipped upwards.

4.3.2 Height experiment 2

During the previous experiment it was observed that the flame covered a larger area of the pot when it was moved further away. For this reason it seemed reasonable that the height would have a bigger impact on a larger pot. A second experiment using the Trangia 25, which is the larger TSK, was therefore conducted.

Since it was observed in Height experiment 1 that covering the air holes might affect the flame, the method was altered during the second experiment. To raise the pot, the UWS stayed in the normal position, and spacers were placed under the pot. The new position was 25 millimetre higher than the current one. Only two positions were examined during this experiment; normal position and raised position. The normal position served as the control. To minimize the differences in starting temperature, the water was taken from a bucket filled at the beginning. Between each test the pot was cooled down in a second bucket to yet again control the initial temperature.

Below is listed the method run for the experiment:

- **Step 1**: Fill bucket with water
- **Step 2**: Fill second bucket with water
- **Step 3**: Light the spirit burner and let it burn for around 4 minutes
- **Step 4**: Fill pot with 300 ml of water
- **Step 5**: Measure initial temperature of the water
- **Step 6**: Put the pot in the TSK
- **Step 8**: Measure time to reach 20 degrees Celsius

- **Step 8**: Measure time to reach 40 degrees Celsius
- **Step 9**: Measure time to reach 60 degrees Celsius
- **Step 10**: Measure time to reach 80 degrees Celsius
- **Step 11**: Remove the pot from the flame and pour out the water
- **Step 12**: Cool the pot in the second bucket of water
- **Step 13**: Repeat steps 4 through 12 two additional times
- **Step 14**: Extinguish the flame and place the spacers in the TSK
- **Step 15**: Repeat steps 3 through 12 two more times

As seen in Figure 12 below the water temperature increased faster when the pot was further away from the flame. It is believed that this was due to the flame covering a larger area of the pot during heating. However, due to the UWS covering the flame it was difficult to get a clear view. However, it was observed that the flame reached around the sides of the pot when in the raised position. Something it did not do when in the lower position.

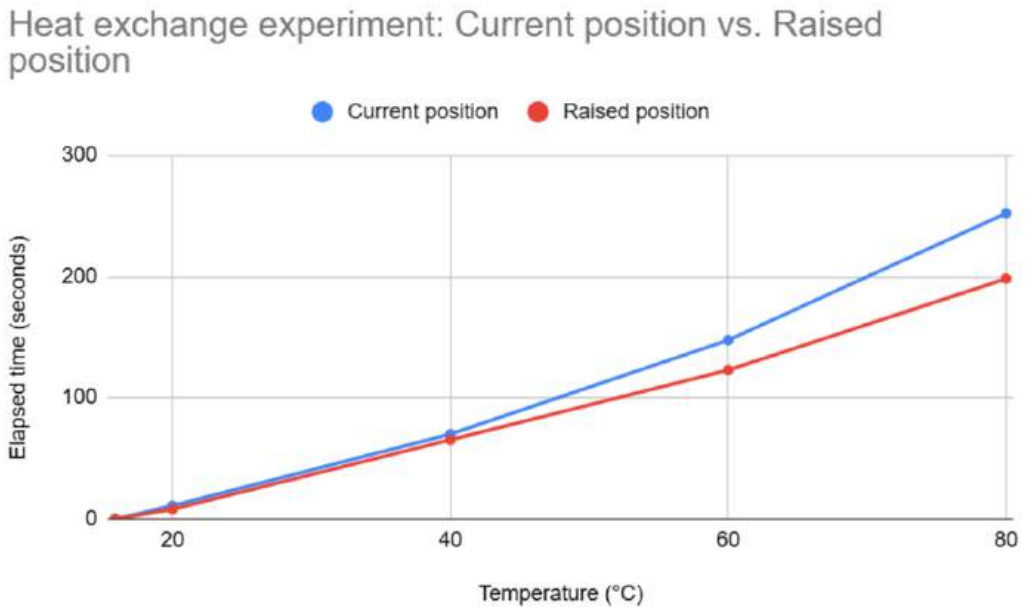


Figure 12: *Results from Height experiment 2*

As opposed to the previous experiment, the UWS and LWS were unchanged during testing, thus giving a more realistic result. Although, the burner is believed to have improved over the course of the experiment, as it had not been used much beforehand. This could have affected the outcome of the test, but it is difficult to say by how much.

These results do show that there is likely a difference in heat transfer depending on the distance between burner and pot. On average, the difference to reach 80 °C

was 54 seconds, with the increased height being the faster. Since this experiment only tested one additional height, with the intention to see if there was a noticeable difference, there is cause to try and find an optimal position.

It is also important to note that it has not been tested how different heights affect the heating capacity when using a gas burner. As the TSK can also come equipped with a gas burner, it is important to control how the height affects this use case.

4.4 Physical research

To have a better understanding of the competition, a store visit was conducted. Within Naturkompaniet as seen in Figure 13 below a store assistant presented several alternatives to the Trangia Storm Kitchen from Primus, MSR and Biolite.



Figure 13: *Store assistant holding the Primus Storm kitchen*

For the Primus product, the container supports rely on individual hinges to rotate between the two positions. According to the assistant, this is the case for the majority of their products. It can be considered that Primus is Trangia's main competitor, as another legacy brand of Swedish outdoor gear. The shop assistant also mentioned that the majority of users for all storm kitchens purchase the gas rather than spirit burner option. The assistant also argued that Trangia's robustness can be partially attributed to using rotating locking solutions (no loose parts).

The in-person product research was continued at Alewalds and other outdoor stores on Kungsgatan. Below are listed the key discoveries from analysing with the related images below:

- **1**: For cheap versions found in Alewalds, fragile flimsy welds were used, along with rivets, this can be seen in the sketch Figure 14c.

- **2**: On several examples of storm kitchens, rivets were used though with varying levels of sturdiness and success, seen in both Figure 14b and 17a.
- **3**: It was noted that for several examples the rim of the wind shield could be used as container support rather than dedicated supports as seen on Figure 14d.
- **4**: Used methods: spot welding, welding, rivets, screws, hinge



(a) *Windmaster* (b) *MSR* (c) *Stabilotherm* (d) *Primus*

Figure 14: Comparison of different burners

4.5 Online Research

An online study was performed by finding information about Trangia products on outdoor forums and YouTube. The goal of this study was to gain further knowledge about the product usability.

4.5.1 Competitors

Listed below are the most common competitors to the Trangia Storm kitchen and the brands offering as a whole. Several of these brands are common places in Sweden, others more international.

In Table 6 below is a collection of the prices for varying versions of the TSK. This can be used as reference against the competitor products.

Table 6: Table of the TSK options with all choices of materials, no Kettle

No.	Type	price (SEK)
1	27, Spirit Burner	792-1595
2	27, Gas burner	1520-2358
3	25, Spirit burner	869-1817
4	25, Gas burner	1603-2594

Biolite:

Biolite is from NYC in the US. Their innovative offering to the outdoor stove space is the CampStove 2+ (CS2). The CS2 burns woodland sticks for fuel and has a USB port to power a phone [7]. The product is 1600 SEK for just the stove and no pots; an expensive offering. The expected users could be interpreted as amateur-outdoor

people which rely on and enjoy new technology. For this reason Biolite is not a direct competitor. An image of the campstove can be seen in figure 15 below.



Figure 15: *The Biolite CampStove 2+*

country of origin, price for alternative, what is the product offering, expected users, innovative/traditional, are they a direct competitor

Esbit:

Esbit is a German manufacturer of camping equipment founded in 1949. They make products similar to Trangia; stackable cooksets mainly focused around spirit or solid fuel burners. Their product offering also includes drinking bottles, food jugs and cutlery [8]. An Esbit stove is sold at a similar price to that of a TSK, providing the same equipment [9]. However, Esbit offers a more modern expression, using for example heat exchangers in their spirit cooksets. The company can be seen as a direct competitor to Trangia, although not as common on the Swedish market. In Figure 16 below is an image of an Esbit cookset using an alcohol burner with a similar design to the Trangia burner.



Figure 16: *The Esbit cookset with alcohol burner and heat exchanger*

Firemaple:

Firemaple is from south east china, though the website tries to appear as a US run company [10]. The brand offers a mid level product at a low price. Their packable cookset is called the cooking kit, which can be seen in figure 17 below. The kit is sold at a similar price to the Trangia with similar function. The kit however packs poorly, with the parts stacked, but loose. The company can be seen as a direct competitor to the TSK with a similar designed gas burner to Trangia's offering and similar price. The expected users are new to outdoor cooking and are willing trial new brands.



(a) *Gas burner* (b) *Stacked cooking kit*

Figure 17: The Firemaple cookset

Jetboil:

Jetboil is an American manufacturer of lightweight, gas powered camping stoves. Unlike Trangia, Jetboil emphasizes cooking speed and lightness as their main selling point. Their focus is largely on top-mounted gas burners, although they offer more traditional gas stoves in their "Basecamp" range [11]. Consequently, Jetboil caters to

hikers and backpackers wanting to cover long distances. For this reason, Trangia and Jetboil are not in direct conflict on the market. A corresponding Jetboil product to the TSK retails at around 1450 SEK, which is a slightly higher price [12]. However, the TSK would include an additional pot, as well as more sufficient wind sheltering and stacking options when packing. Additionally, the bundle could not be found in any Swedish shops. To assemble the bundle from its individual counterparts would cost almost 2500 SEK [13] [14]. In Figure 18 below is a picture of the MightyMo Group Cook Bundle, deemed to be comparable to a TSK.



Figure 18: *The MightyMo Group Cook Bundle from Jetboil*

MSR:

Mountain safety research (MSR) is a function focused company based in the US. The products are high quality designed for serious outdoor people and professionals. Their cooking products are only a small part of their offering. They also offer tents, poles and general mountain tools. The MSR alternative to the TSK is the Windburner stove system combo at 3500 SEK, see figure 19 below. The materials are specialised with a ceramic coated aluminium pot and pan. The pot can also hold up to 2,5L is sufficient for a whole group [15]. MSR would not be considered a direct competitor as MSR would be a large purchase and not the same price range.



Figure 19: *The Windburner Stove system combo from MSR*

Primus:

Primus can be considered as Trangia's most direct competitor. The brand is from Stockholm and has a similarly high level of heritage in Storm kitchens to Trangia. The alternative to the TSK is called the Primetech stove set 1.3L, which can be seen in figure 20 below [16]. The product costs 1500 SEK [17].



Figure 20: *The Windburner Stove system combo from MSR*

4.5.2 Reviews

By searching *Trangia kitchen reviews* on Google, several short articles were found. The focus during the search was to find common points from different users. Reviews from four different long-time users were examined.

One of the main takeaways from the reviews was that everyone agreed on the durability of the TSK as one of its main selling points. This goes together with the simplicity of the TSK, and that there are very few pieces that can break. One of the reviewers said "There is nothing to go wrong - throw them over a cliff and collect the bits at the bottom and cook your dinner." [18]. All of the reviewers had used

the spirit burner and found the ease of finding fuel to be one of the more important factors [19]. The compact way the TSK packs together was additionally one of the more important factors for buying a storm kitchen from Trangia. One reviewer mentioned that the the clever storage design is one of the main reasons for buying a TSK [20].

Not many disadvantages were found for the TSK. The one common problem, that all reviewers agreed on, was the difficulty to use the simmer ring. One reviewer described the process as finicky [21]. No problems were mentioned about either the CS or the wind shields.

4.5.3 Videos

On Youtube, several reviews can be found which highlight the advantages and limitations of the TSK.

A brilliant source to study the development of the TSK has been a video called '1980's vs. 2022 Trangia Comparison - what's changed?'. It is remarkable how little has changed with the TSK, the CS after all are the same. The sparring however has gained the rotating cover over time and is placed rather than in 80's when it was on a hinge [22]. A screenshot from the video can be seen in Figure 21 below.



Figure 21: *Screenshot from the comparison video*

Another video of value is 'Headspace Performance Test / Alcohol Stove / Spirit Burner / Gen2 Nano'. Within this video, the distance of the flame from a container is analysed. The video tests the time for water to boil and the amount of fuel used. This is when the Trangia spirit burner is used with the Firebox nano, in its three different heights [23].

Though the test was quite rudimentary, the concept of varying heights could be

tested to enhance any future CS design. A screenshot from the video can be seen in Figure 22 below.



Figure 22: *Screenshot from the firebox experiment video*

5 Branding

Within this chapter is a short background on the Trangia brand Identity.

5.1 Brand Platform

Though the potential design change is not considered an extension, a brand platform has however been made. This is to assess the brand identity for possible design changes.

This brand platform focuses on the materials and form of the TSK and physical components. Colours and packaging have been deemed outside of the scope.

The information below has been derived from Trangia's internal product design guideline.

Heritage

Trangia is currently celebrating its 100th year anniversary. It is considered a brand that presents a traditional choice for outdoor cooking.

Vision

Trangia's vision is to create Swedish made products which promote good values in sustainable living and design. The materials used are simple and they aim to create robust long lasting products.

Mission

Trangia's aim is to 'grow interest for nature' through good design and passion. This should lead to customer satisfaction and an emotional connection to the product.

Values

Trangia's core values are presented in three key words: 'Simplicity, Quality and reliability'.

- Simplicity - Clean, simple lines and soft shapes. A style that should work for over a hundred years
- Quality - Modularity-guaranteeing longevity. Ensuring high quality, included with climate conscious material choices
- Reliability - The products should function in all situations

Offer

Trangia's offering is access to nature through reliable and sustainable products for everyone.

Audience

Trangia resonates with those interested in nature and the outdoors. The audience prefer reliability over technical advancements.

Benefits

By prioritising reliability, through quality products and design, Trangia creates simplistic products that function well. Users benefit from a product that can be considered a long term investment.

Personality

Trangia's personality can be considered as no thrills design. The products are simple with function being the leading factor.

Product form

Trangia's overall silhouette is integral to their brand identity. This can most clearly be seen in Figure 23 below, showing the Trangia Logo. The logo shows the design elements needed to ensure any change is congruent to the brand's identity: Broad form, simple lines and robust clear components.



Figure 23: *Trangia logo and script (Trangia)*

5.2 Design Expression

To exemplify the Trangia design expression, several 'This-Not that' image boards were produced. These were defined through three concept areas: Details, Expression and complexity. The image boards use images of existing designs to exemplify how Trangia develop a design and what characteristics are required for any new design to be congruent.

Details

The details expression image board in Figure 24 highlights the specific features of a Trangia design. For the example a Trangia designed radio would use patterned circles as apertures rather than thick bold lines. Additionally the edges on a Trangia designed product are soft and highly filleted. This avoid the design communicating aggression or speed like the mouse and car headlight on the 'Not that' side.

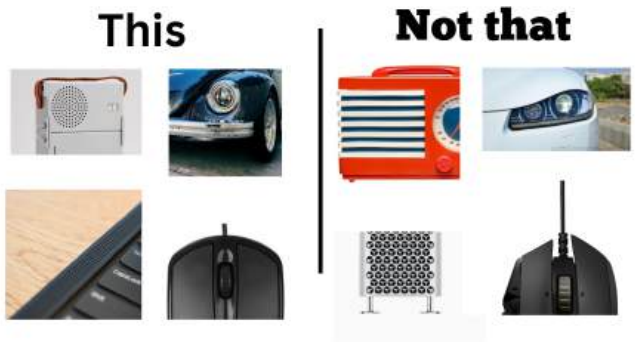


Figure 24: *A visualization of suitable design details for Trangia products*

Expression

The expression image board, seen in Figure 25 below, shows the same types of products but with different design expressions. The products shown on the left have a similar expression to that of Trangia's products. These designs can be described as minimalist and robust, using proven functions and features. For example, the wooden hammer uses traditional materials such as wood and steel. There is essentially no need to change the design of this product, unless to 'modernize' it. It's counterpart, shown to the right, uses novel materials for the same purpose. This gives it a sense of modernity and a more aggressive look, an expression that is not in line with Trangia as a brand.



Figure 25: *A visualization of the Trangia design expression*

Complexity

The Complexity image board in Figure 26 shows images on the left that are minimalistic and practical. The products on the right are complex with features rather than practicality as the focus. For instance, the Lundhags boot uses minimal materials and joining lines. The Salomon trail shoe in contrast has several technical materials, and unusual shoe structure. While both shoes are high quality and appropriate for hiking the Lundhags follows similar design principles to Trangia's products.



Figure 26: *A visualization of the complexity communicated through Trangia's products*

5.3 Features and design elements

An additional part of the brand identity for consideration are the features and elements used in Trangia's designs. These parts are presented below.

Holes

For both the Tamoj and the LWS, small holes are used in lines as a functional design element. The function for the Tamoj is mainly for weight reduction, and the brands style is its main purpose. An image of the Tamoj can be seen in Figure 27 below. For the LWS 4 rows of holes are used for ventilation, but are also a brand identifier.



Figure 27: *Picture of the Trangia Tamoj*

Fixings and attachments

Trangia uses several variations of fixings and attachments. Presented in Figure 28 below are several of them.



(a) *Trangia gas burner part secured*



(b) *The Trangia 25-strap*



(c) *The Trangia T-cup*

Figure 28: Example Trangia products (Trangia.se)

Listed below are similar aspects of the components shown above that should be adhered to for any potential CS design:

- Materials: Steel, Aluminium, Brass, Hard rubber, HDPE
- Surface finish: Brushed, cast, unpolished, matt
- Colours: Black, silver, red, yellow, navy
- Form: Small parts are chunky and have a high mass. Larger parts have high surface area to volume and weight. All components have a low center of gravity.

5.4 Brand positioning map

Through a Brand positioning map, a brand can be better understood and direct product development. This was done based on the research described above. The brand positioning map can be seen in Figure 29 below.

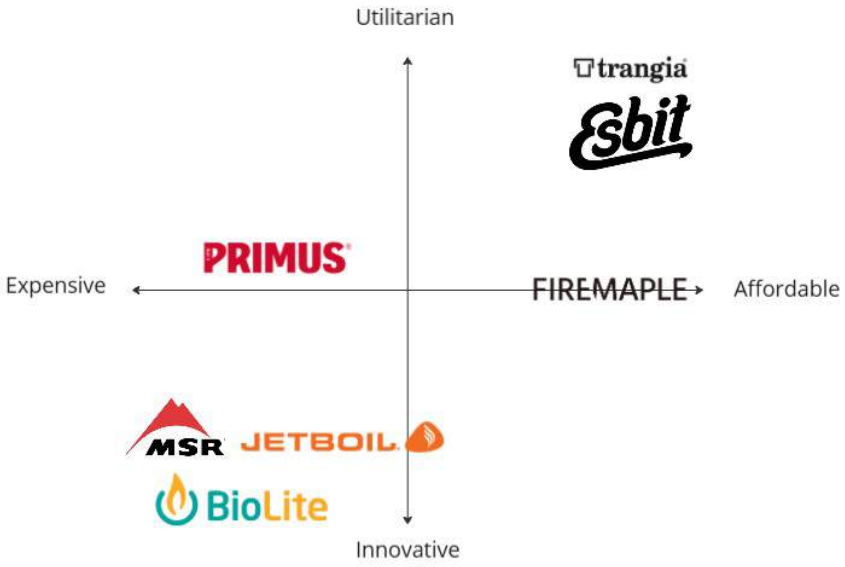


Figure 29: *Trangia Brand positioning map with storm kitchen competition brands*

With any design proposals, the brand position map, shows that the overall design must continue to be utilitarian in purpose and continue to be affordable. This will ensure that the Trangia Storm kitchen holds its market position.

6 User study

Following human centred design practices it is deemed appropriate to first understand the user needs and current issues. This would prevent ideating in a way that does not meet user requirements.

Within this section these user requirements are found through a Survey. As the product development is focused on manufacture, less focus is put on user studies.

6.1 Survey on user experience

A survey was created in both English and Swedish with the links posted in relevant Reddit groups and forums online; This included the DofE reddit group, UK hill walking forum amongst others listed in appendix A, the Swedish sites are also similarly listed. Due to rulings on Utsidan and other sites the Swedish version of

the survey only received 15 answers. The English version, however, received 140 answers. This was deemed to be a very reasonable sample size.

The survey was comprised of 10 questions, both multiple choice and text-answer. These questions were written to ascertain the general users opinion of the TSK as a whole, and additionally the CS and Sparring. The questions are shown in the list below. To view the survey in further detail, see Appendix A.

- **1**: Have you used a Trangia storm kitchen?
- **2**: Have you ever had issues with the containers' supports on the Trangia Storm kitchen (i.e. longevity durability)?
- **3**: Do you feel other brands have a better solution for holding different sized containers?
- **4**: Do you use a gas or a spirit burner?
- **5**: Why do you prefer using a gas burner?
- **6**: Why do you use a spirit burner?
- **7**: Do you use the Trangia simmer ring to control the flame?
- **8**: Why do or don't you use the simmer feature?
- **9**: Do you feel other brands have a more user friendly solution for simmer cooking?
- **10**: Do you have any other overall thoughts for improvement?

6.1.1 Survey results

Using research methodology practices the results of the survey were collated into key themes for each question when relevant. For this process Miro was used.

In Figure 30 below it can be noted that the majority of users have no issue with the container supports (126/140). The only issues stated were: chance of bending of the CS, tolerance issues, and sharp edges.



Figure 30: *Miro board screenshot of collated point for CS with quotes included*

For the question ' Do you feel other brands have a better solution for holding different sized containers?' , it can be seen in Figure 31 that most users were unaware of any better solutions. The only other consideration was that other brands allowed users to use a coffee pot on top of the stove.



Figure 31: *Miro board screenshot of collated point for competition Brand solutions with quotes included*

To the left in Figure 32 below, it shows that the majority of users on the English speaking surveys used the spirit burner. This could likely be down to the choice of forums, specifically the DofE group. In contrast, for the Swedish survey only 4 of the 15 users had ever used a spirit burner and not as their only source of heat when cooking with the TSK. This can be seen to the right in Figure 32 below.



Figure 32: *Miro board screenshot of rate between spirit and Gas use*

For the next question it can be seen in Figure 33 , that gas users preferred it due to the cleanliness, ease of temperature control and weight. It can be noted however that Spirit users found gas to be inconvenient: it is hard to ascertain how full the gas is, and harder to access. Spirit appears to also have a following due to it being the more traditional means for outdoor cooking.



Figure 33: *Miro board screenshot of preferences for gas or spirit burners*

The next image below, denotes why users use the simmer ring. A key positive is the ability to reduce fuel waste. The component also limits heat and prevents food from burning. Users however complain that the flame is lacking with the simmer ring but far too large without it. The most common use case, is simply to extinguish the flame. As the lid has a gasket that can melt this use case is a necessity.



Figure 34: *Miro board screenshot of opinions on the simmer ring*

7 Trangia Factory visit

The Trangia factory was visited to better understand the manufacturing processes. Discussion with the Trangia supervisor also assisted in refining the scope of the project and providing key information.

During the visit at the Trangia factory a tour of the machines and manufacturing techniques was shown. Thanks to the choice of day, the majority of the machines used for the manufacture of the UWS were in use.This was of value to understand what potential manufacturing methods a new design could use.

7.1 Manufacturing techniques within Trangia's factory

Shown in Table 7 below is the manufacturing techniques presented at the Trangia factory.

Table 7: Table for the Trangia manufacturing techniques

Number	**Technique**	**Use case**	**Labour input**
1	Deep drawing	Shaping blanks into pots and containers	Fully automated
2	Wire bending	Bend straight wire into a kettle handle	Fully automated
3	Clinching	Pressing the handles onto the food containers	Worker controlled
4	Riveting	Fixing handle on to the kettle	Worker controlled
5	Blanking	Creating the Kettle spout hole	Worker controlled
6	Die cutting	On LWS for air holes	Worker controlled
7	Flange bending	On edge of the wind shields	Fully automated
8	Metal Stamping	For the CS, must also be tumbled	Fully automated
9	Wire pressing	Add flat edge to hold the CS	Worker controlled
10	Clamping	Pressing CS to the ring	Worker controlled
11	Press stamp-cut	Creating the strap holders	Fully automated

These techniques are all used during the process of manufacture for the TSK. There are several components purchased externally, particularly for the gas burners.

Shown in Figure 35 below is deep drawing; 'Technique 1'. Where a blank sheet is held in a die and then drawn by the punch into the container shape.

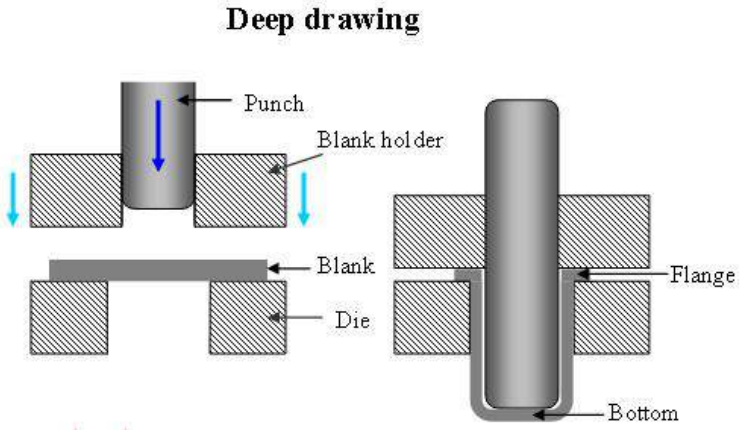


Figure 35: *Technique 1- deep drawing (Metallurgy for dummies)*

'Technique 2' is presented in Figure 36 below. The wire is bent by a rotating die which presses at varying angles into the form of the handle.



Figure 36: *Autonomous wire bending (Clark Engineering)*

'Technique 3' can be seen below in Figure 37. This is an excellent process, requiring no additional fixings. The added part is forced behind the sheet, creating lips that secure it from the inside. The connection is watertight.

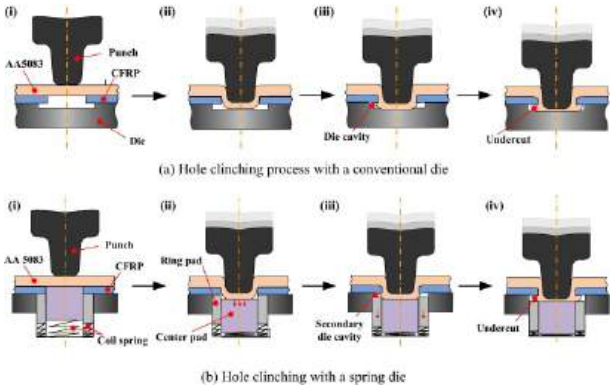


Figure 37: *Clinching process (Science direct)*

Figure 38 below presents how riveting; 'technique 4' is done. Where two sheets of metal are joined through a pin that is fixed by its two ends pressed against thee sheets outer walls.

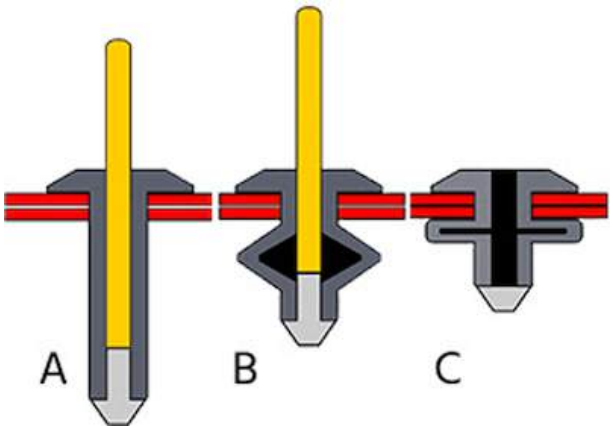


Figure 38: *Riveting process (tameson.com)*

For 'technique 5' a hole is created in the sheet metal with a blank punch. This can be seen in Figure 39 underneath.

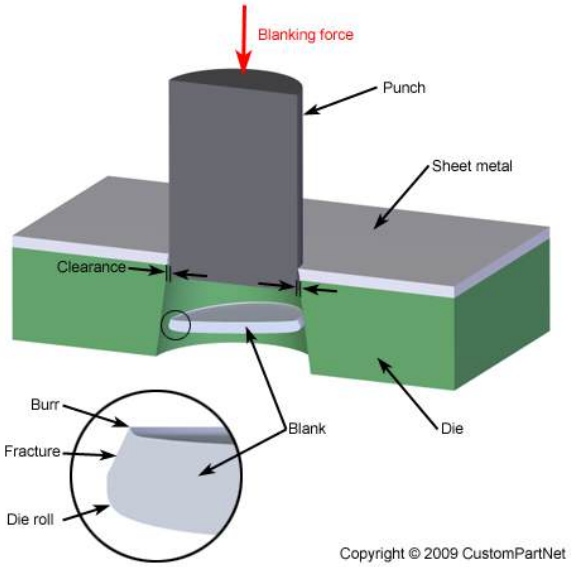


Figure 39: *Blanking process (Global Spec)*

Die cutting, 'technique 6' is a very common process in manufacturing for metal sheets. To create a hole, the die will have a sharp perimeter cutting and removing the material, this can be seen in Figure 40 below.

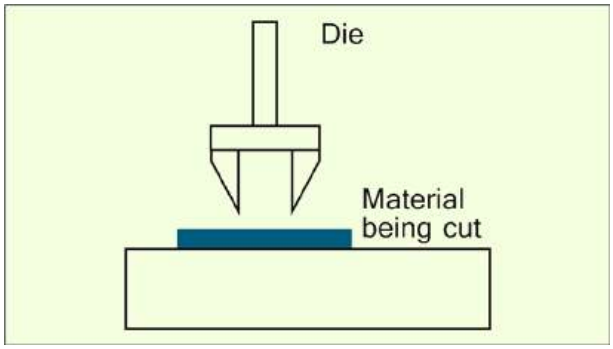


Figure 40: *Die cutting process (Accu-shape Die cutting)*

For lack of a clear name, 'Technique 7' has been called flange bending. It is a process where the edge of the metal is bent backwards. In Figure 41 the process is shown for only one flat edge. At the Trangia factory, the UWS amongst other parts have their flanges bent back with a rotating punch around the edge of the containers.



Figure 41: *Flange bending process (HLH rapid)*

In Figure 42 below, 'technique 8' , metal stamping is shown. The sheet metal is pressed between a punch and die, forming the metal.

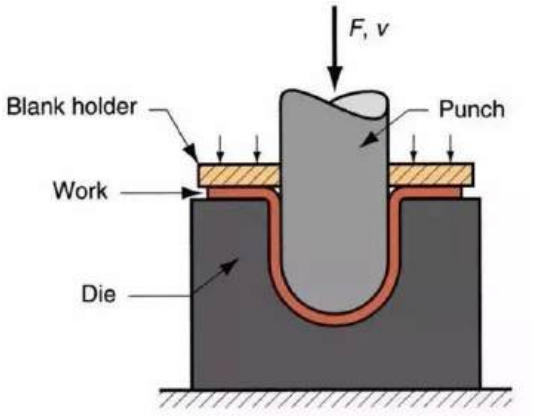


Figure 42: *Metal stamping process (Sinoway Industries)*

'Technique 9' differs from 'technique 2' by being manually driven. As seen in Figure 43 the worker at Trangia, places the ring and then starts the machine. The ring is then pushed in at 3 spots. This processes follows the same principles as the wire bending.



Figure 43: *Wire pressing process (Trangia-Factory)*

Clamping, 'technique 10' is shown in Figure 44 underneath. The process involves clamping the CS around the ring in the spots previously created by 'technique 9'. It is the industrial equivalent to using a pair of pliers pressing metal together.



Figure 44: *clamping process (Trangia-Factory)*

The stages of the manufacture as described above can be seen in figure 45 below.



Figure 45: *Stages of manufacture for the current hooks and Ring*

'Technique 11' is press stamp-cut. This is a combination of technique 8 and 5 and is a fully autonomous process.

7.1.1 Key issues for the manufacture of the CS and ring

During Technique 8, as described in Table 7, it has been found that the process is relatively simple and fully automated. The key issue that has led to the need for this project has been related to the ring. It has been found that 'Technique 10' is difficult due to tolerance issues with the ring. If the ring is too short or long it's not capable of following its function. The incorrect length can lead to entire batches, provided by the supplier, being thrown as they are unusable.

In regards to Technique 9, if the wire pressing is not adequate, then the CS can slide and not function properly. The depth of the bend also affects the length of the ring, and has to be monitored to keep it within specification. Issues with this leads to waste material, with failed attempts thrown alongside the rings that have the wrong length.

For an improved manufacturing process, it must reduce the amount of wasted material. The process must also limit worker interaction, reducing the time spent correcting issues.

7.1.2 Cost of manufacture

The cost of manufacture has been determined by Trangia, and the information will be a limiting factor in the specification. This information has been shared but however has to be omitted from the report due to its sensitivity.

7.1.3 Pace of manufacture

The pace of manufacture has been explained and noted to everyone involved within this project. The information will directly influence how changes to the design will be approached, the information however is unavailable to the public.

7.1.4 Materials used

Trangia predominantly uses four different material options: ultralight aluminium, hard anodized aluminium, non-stick and Duossal [4]. These materials can be seen in figure 46 below.



Figure 46: *Pots in the four different materials. From the top: Ultralight aluminium, hard anodized aluminium, non-stick, duossal*

The wind shield, both upper and lower are manufactured in aluminium. The pots and pans however use these specialised materials.

7.1.5 Alternative Materials

For any solution it could be possible to use differing materials other than Trangias current choice. Within the literature study, materials were researched through Edupack. Specialist materials could enhance the TSK. It has been deemed, however, that this would be non-congruent to Trangia's brand. Trangia uses reliable common materials, and any change to the design should continue to use materials previously found in their catalogue of products. This additionally would prevent any logistical issues, as any current material can easily be sourced through their current supply chain.

8 Further experiments

After the Trangia factory visit, it was determined that further testing was needed to validate the current design. Foundational elements of the design needed to be confirmed, such as air supply and further height variations. Experiments were conducted for both the gas burner and the spirit burner to find potential improvements.

8.1 Air flow experiment

Discussing with the Trangia supervisor, it was ascertained that there was no record for the choice of size and number of air holes on the LWS. This is likely due to the age of the design. It was also suspected that restricting the airflow affected the results of the first experiment. Because of this, it was deemed reasonable to test a different rate of air supply as it could increase the flame efficiency on the spirit burner.

To test this, an experiment and method was created. By reducing the number of

holes on the top and side of the LWS, air supply entering and exiting could be controlled. In Figure 47 an example illustration is shown.

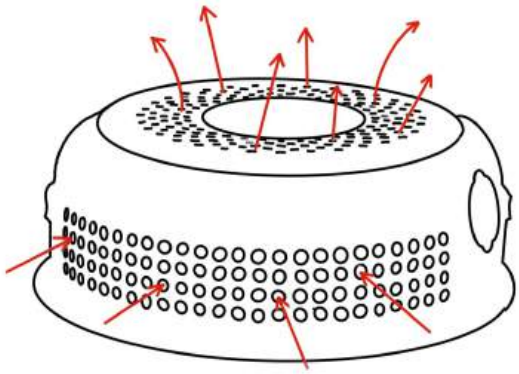


Figure 47: *Illustration of air flow path on the LWS*

To affect air flow, electrical tape was used, blocking percentage amounts of the holes; an example of this from the experiment is shown in Figure 48 below. This image presents when the side is fully blocked; 0 percent air flow side.The top of the LWS has 3 out of the 5 rows of holes open; 60 percent air flow.



Figure 48: *Image of taped air flow path on the LWS*

Within Figure 49 below are the different combinations used in the experiment to test the air flow.

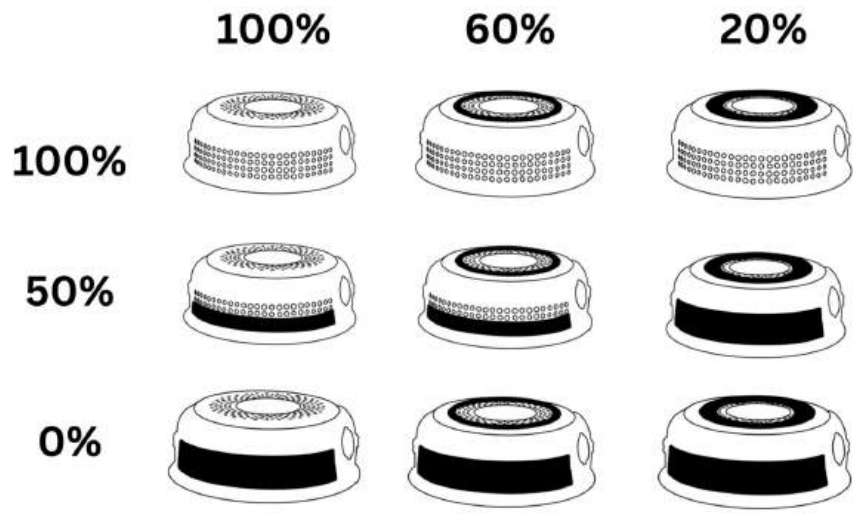


Figure 49: *Combinations of tape variation between the top and side of the LWS*

Listed below is the method run for the experiment:

- **Step 1**: Fill bucket with water
- **Step 2**: Fill second bucket with water
- **Step 3**: Light the spirit burner and let it burn for around 4 minutes
- **Step 4**: Fill pot with 300 millilitres of water
- **Step 5**: Measure initial temperature of the water
- **Step 6**: Put the pot in the TSK
- **Step 8**: Measure time to reach 20 degrees Celsius
- **Step 8**: Measure time to reach 40 degrees Celsius
- **Step 9**: Measure time to reach 60 degrees Celsius
- **Step 10**: Measure time to reach 80 degrees Celsius
- **Step 11**: Remove the pot from the flame and pour out the water
- **Step 12**: Cool the pot in the second bucket of water
- **Step 13**: Repeat steps 4 through 12 two additional times
- **Step 14**: Extinguish the flame and then add or remove tape for the next combination of air flow holes.
- **Step 15**: Repeat steps 3 through 12 three times.

Below in Figures 50, 51, 52 are the graphs reflecting the effect that the air flow combinations had on the rate of heat exchange.

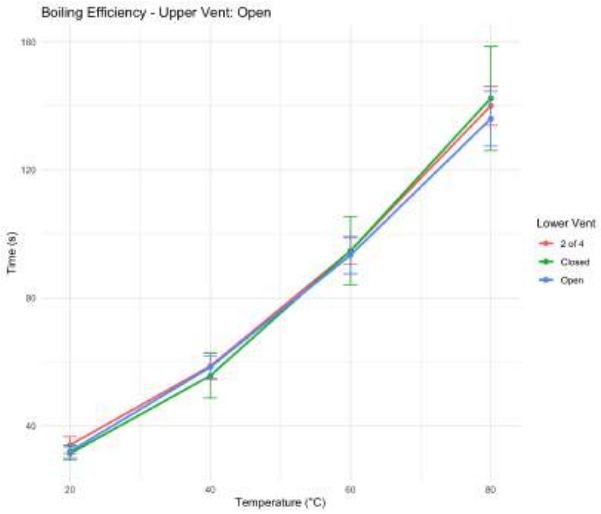


Figure 50: *Results from experiment changing Air flow, graph presents when upper is fully open and lower varied*

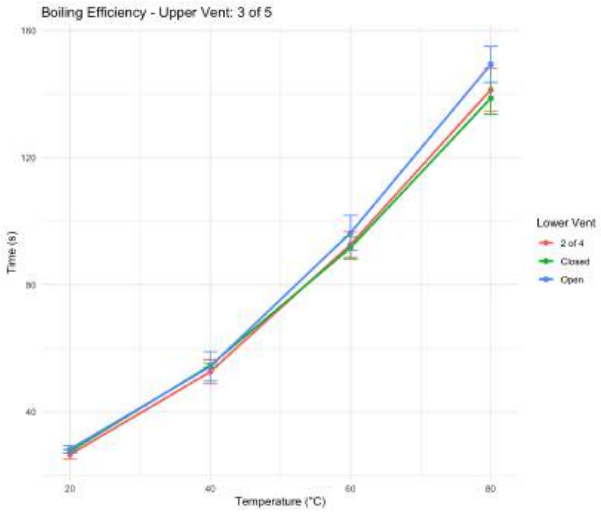


Figure 51: *Results from experiment changing Air flow, graph presents when upper is 60 percent open and lower varied*

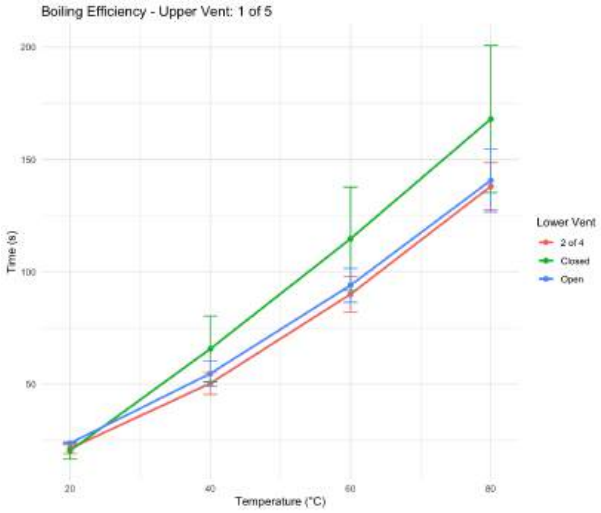


Figure 52: *Results from experiment changing Air flow, graph presents when upper is 20 percent open and lower varied*

The experiment appeared to be inconclusive for whether the air flow largely could be improved by removing holes. It did however appear that removing all of the holes on the lower part of LWS reduced the efficiency of the flame. At 80 degrees the closed lower vent took 17.3 second longer than the open vent.

It could be considered that any reduction to the air vents would not be considered an improvement for efficiency. This experiment does not however, test if more air vents would improve the efficiency.

8.2 Gas burner medium heat, height experiment (300 millilitres)

The effects of pot height when using the gas burner were tested in a series of experiments. The first experiment, which is presented below, used 300 millilitres of water and the time to reach 80 °C was measured. This was done with the pot at the current height, raised 15 millimetres and raised 25 millimetres. The height was set using metal spacers of appropriate height. Each time, the control knob on the burner was opened one full rotation. Figure 53 below presents the set up of the experiment for holding the thermometer.



Figure 53: *Photo taken during the preparation of the experiment*

Listed below is the method run for the experiment:

- **Step 1**: Fill bucket with water
- **Step 2**: Fill second bucket with water
- **Step 3**: Turn the control knob on the gas burner one full turn. Light the gas burner.
- **Step 4**: Fill pot with 300 millilitres of water
- **Step 5**: Measure initial temperature of the water
- **Step 6**: Put the pot in the TSK
- **Step 8**: Measure time to reach 20 degrees Celsius
- **Step 8**: Measure time to reach 40 degrees Celsius
- **Step 9**: Measure time to reach 60 degrees Celsius
- **Step 10**: Measure time to reach 80 degrees Celsius
- **Step 11**: Remove the pot from the flame and pour out the water
- **Step 12**: Cool the pot in the second bucket of water
- **Step 13**: Repeat steps 4 through 12 two additional times
- **Step 14**: Extinguish the flame and then add the spacers to raise the pot.
- **Step 15**: Repeat steps 3 through 12 three times
- **Step 16**: Add the second set of spacers and repeat steps 3 through 12 three more times

The experiment showed that increasing the distance between the spirit burner and the pot also increased the heating time of the water. During the control test, the water reached 80 degrees Celsius within an average of 111,3 seconds. When the distance was increased by 15 and 25 millimetres, the times increased to 131,3 and 133,7 respectively.

These results are shown in Figure 54 below.

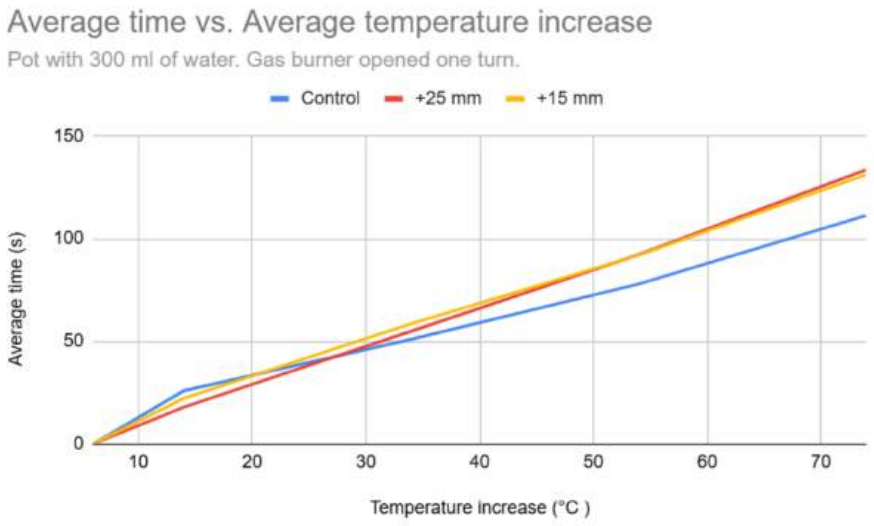


Figure 54: *Results from experiment changing height of pot while heating 300 millilitres of water using gas burner*

8.3 Gas burner medium heat, height experiment (1 Litre)

In the following experiment, a pot filled with 1000 ml of water was tested at two different heights; the current position (control) and an increase of 15 millimetres. Following the previous experiment it was of interest to see how much the heating times would change when using a larger quantity of water. The +25 millimetres position used in the previous experiment was excluded due to its heating time being similar to that of the + 15 millimetres. The supports used can be seen in figure 55 below.



Figure 55: *Photo taken of the two supports used to raise the height of the pots*

Listed below is the method run for the experiment:

- **Step 1**: Fill bucket with water
- **Step 2**: Fill second bucket with water
- **Step 3**: Turn the control knob on the gas burner one full turn. Light the gas burner.
- **Step 4**: Fill pot with 1 litre of water
- **Step 5**: Measure initial temperature of the water
- **Step 6**: Put the pot in the TSK
- **Step 8**: Measure time to reach 20 degrees Celsius
- **Step 8**: Measure time to reach 40 degrees Celsius
- **Step 9**: Measure time to reach 60 degrees Celsius
- **Step 10**: Measure time to reach 80 degrees Celsius
- **Step 11**: Remove the pot from the flame and pour out the water
- **Step 12**: Cool the pot in the second bucket of water
- **Step 13**: Repeat steps 4 through 12 two additional times
- **Step 14**: Extinguish the flame and then add the spacers to raise the pot.
- **Step 15**: Repeat steps 3 through 12 three times

In accordance with the previous experiment, the raised position was slower than the control. On average, the control position reached 80 degrees Celsius in 392,3 seconds, while the raised position required 449 seconds.

The results of the experiment are shown in Figure 56.

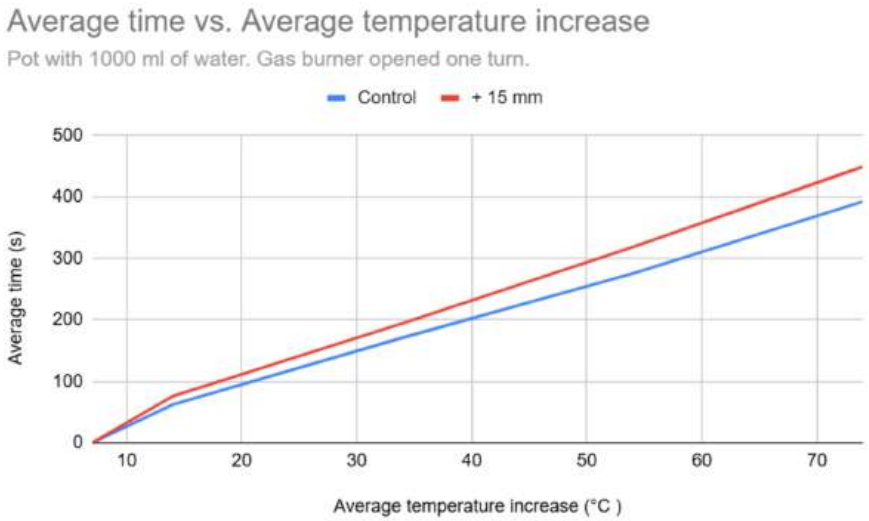


Figure 56: *Results from experiment changing height of pot while heating 1 litre of water using gas burner at medium heat*

8.4 Gas burner 4/3 heat, height experiment (1 Litre)

Half way through the experiments it was discussed if the power output of the flame altered the optimal distance from the burner. An online search was made and a relevant article was discovered. According to [24] the optimal distance for a pot is "70 percent along its length". With this information in mind, the "Gas burner medium heat 1000 ml height experiment" was repeated with the burner at maximum output. At the time, it was impossible to measure the flame, though it was visually confirmed that doing so made the flame longer. Through experimentation, maximum output proved to be at 4/3 turns of the control knob.

Listed below is the method run for the experiment:

- **Step 1**: Fill bucket with water
- **Step 2**: Fill second bucket with water
- **Step 3**: Turn the control knob on the gas burner 4/3 turns. Light the gas burner.
- **Step 4**: Fill pot with 300 millilitres of water
- **Step 5**: Measure initial temperature of the water
- **Step 6**: Put the pot in the TSK

- **Step 8**: Measure time to reach 20 degrees Celsius
- **Step 8**: Measure time to reach 40 degrees Celsius
- **Step 9**: Measure time to reach 60 degrees Celsius
- **Step 10**: Measure time to reach 80 degrees Celsius
- **Step 11**: Remove the pot from the flame and pour out the water
- **Step 12**: Cool the pot in the second bucket of water
- **Step 13**: Repeat steps 4 through 12 two additional times
- **Step 14**: Extinguish the flame and then add the spacers to raise the pot.
- **Step 15**: Repeat steps 3 through 12 three times

Increasing the power output from the burner proved to minimise the difference between the tested heights. However, the control was still the fastest. The average time for the control was 345 seconds, while the higher position took 369,33 seconds. The results from the experiment are presented in Figure 57 below.

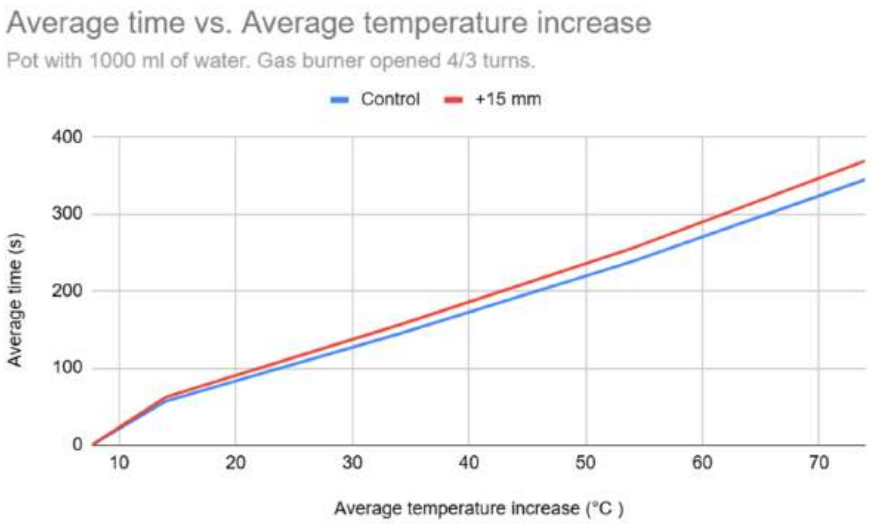


Figure 57: *Results from experiment changing height of pot while heating 1 Litre of water using gas burner at full power*

8.5 Spirit burner height experiment (1 litre)

To accurately compare the efficiency between the gas and spirit burner, an experiment on speed of heating 1000 ml of water was conducted using a spirit burner. This experiment would be far easier to use as a reference rather than any results previously made with only 300 ml of water.

Listed below is the method run for the experiment:

- **Step 1**: Fill bucket with water
- **Step 2**: Fill second bucket with water
- **Step 3**: Ignite the spirit burner. After it has been primed, let it burn for three minutes.
- **Step 4**: Fill pot with 1 litre of water
- **Step 5**: Measure initial temperature of the water
- **Step 6**: Put the pot in the TSK
- **Step 8**: Measure time to reach 20 degrees Celsius
- **Step 8**: Measure time to reach 40 degrees Celsius
- **Step 9**: Measure time to reach 60 degrees Celsius
- **Step 10**: Measure time to reach 80 degrees Celsius
- **Step 11**: Remove the pot from the flame and pour out the water
- **Step 12**: Cool the pot in the second bucket of water
- **Step 13**: Extinguish the flame
- **Step 14** Add the spacers to raise the pot
- **Step 14**: Repeat steps 3 through 13
- **Step 15**: Repeat steps 3 through 14 two times

The spirit burner was evidently weaker than the gas burner. The average control time to reach 80 degrees was 800,67 seconds. Impressively, however, the increased height reduced the time to reach 80 degrees down to 525,67 seconds. That is a reduction of 275 seconds. The results from the experiment are presented in Figure 58 below.

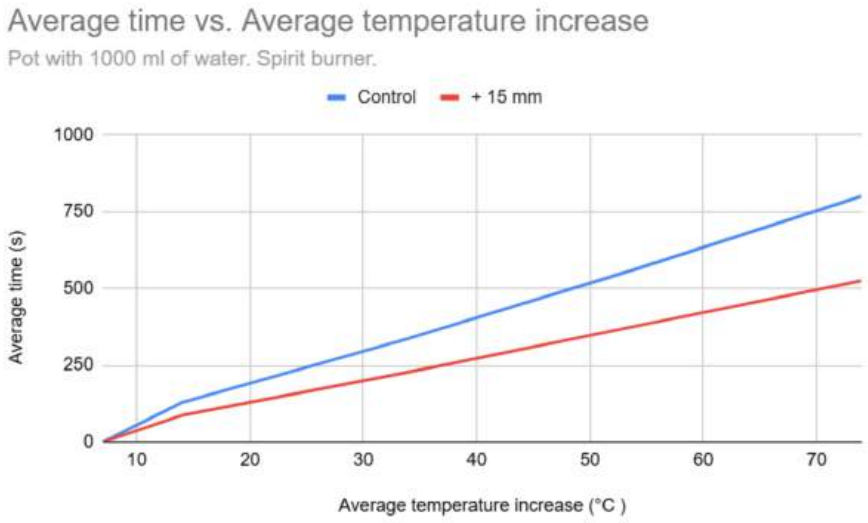


Figure 58: *Results from experiment changing height of pot while heating 1 litre of water using spirit burner*

8.6 Gas burner pan experiment (300 millilitres)

It was realised that any change to the height of the pots could also be made to the pan. The pan is currently 35 millimetres above the ring, see figure 7. In the experiment, the smaller pan from TSK 27 was used within TSK 25, held at the ridge of the CS, 20 millimetres lower, this can be seen in Figure 59 below. The heating element used was the Gas burner.



Figure 59: *The TSK 27 pan lodged against the CS of the TSK 25 during the experiment*

Listed below is the method run for the experiment:

- **Step 1**: Fill bucket with water
- **Step 2**: Fill second bucket with water
- **Step 3**: Turn the control knob on the gas burner one full turn. Light the gas burner.
- **Step 4**: Fill pan with 300 millilitres of water
- **Step 5**: Measure initial temperature of the water
- **Step 6**: Put the pan in the TSK
- **Step 8**: Measure time to reach 20 degrees Celsius
- **Step 8**: Measure time to reach 40 degrees Celsius
- **Step 9**: Measure time to reach 60 degrees Celsius
- **Step 10**: Measure time to reach 80 degrees Celsius
- **Step 11**: Remove the pan from the flame and pour out the water
- **Step 12**: Cool the pan in the second bucket of water
- **Step 13**: Repeat steps 4 through 12 two additional times
- **Step 14**: Repeat steps 4 though 12 with the pan placed in the lowered position

There was a large improvement to the efficiency, cooking with the pan at a lower height by 20 mm. The control time was 50,67 seconds slower at 162 seconds. This was compared against the minus 20 millimetres test at 109,33 seconds. The results from the experiment are presented in Figure 60 below.

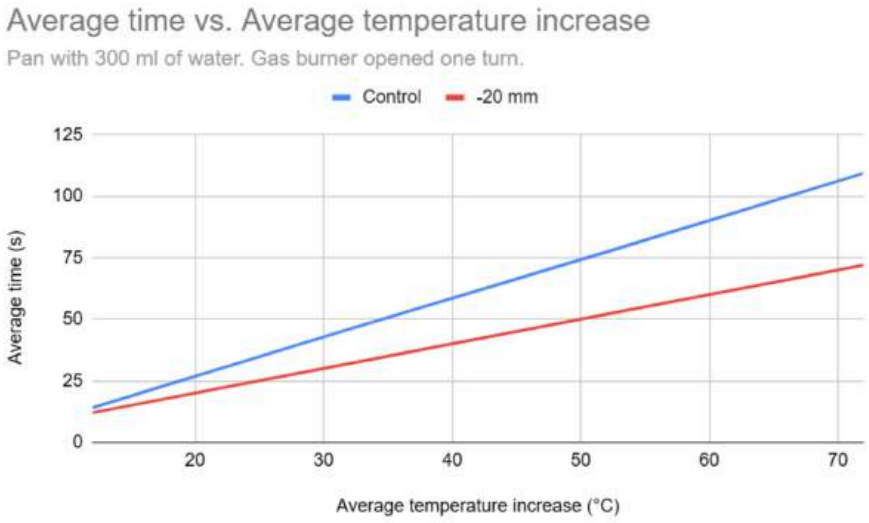


Figure 60: *Results from experiment changing height of pan while heating 300 ml of water using gas burner*

8.7 Spirit burner pan height experiment (300 millilitres)

To compare between the spirit burner and Gas burner, the previous pan experiment was repeated but using the spirit burner instead.

Listed below is the method run for the experiment:

- **Step 1**: Fill bucket with water
- **Step 2**: Fill second bucket with water
- **Step 3**: Ignite the spirit burner. After it has been primed, let it burn for three minutes.
- **Step 4**: Fill the pan with 300 millilitres of water
- **Step 5**: Measure initial temperature of the water
- **Step 5**: Place the adaptors for the small pan on the CS
- **Step 6**: Put the pot in the TSK
- **Step 8**: Measure time to reach 20 degrees Celsius
- **Step 8**: Measure time to reach 40 degrees Celsius
- **Step 9**: Measure time to reach 60 degrees Celsius
- **Step 10**: Measure time to reach 80 degrees Celsius
- **Step 11**: Remove the pot from the flame and pour out the water
- **Step 12**: Cool the pot in the second bucket of water
- **Step 13**: Extinguish the flame

- **Step 14** Remove the adaptors from the TSK
- **Step 14**: Repeat steps 4 through 13 (skip Step 5)
- **Step 15**: Repeat steps 4 through 14 two times

Similarly to the gas burner, there was a large improvement in the efficiency of the spirit burner at a lower height for the pan. The average time for the control was 195,67 seconds, while the lower position at minus 20 millimetres took only 136,67 seconds to reach 80 degrees. The results of the experiment can be found in figure 61

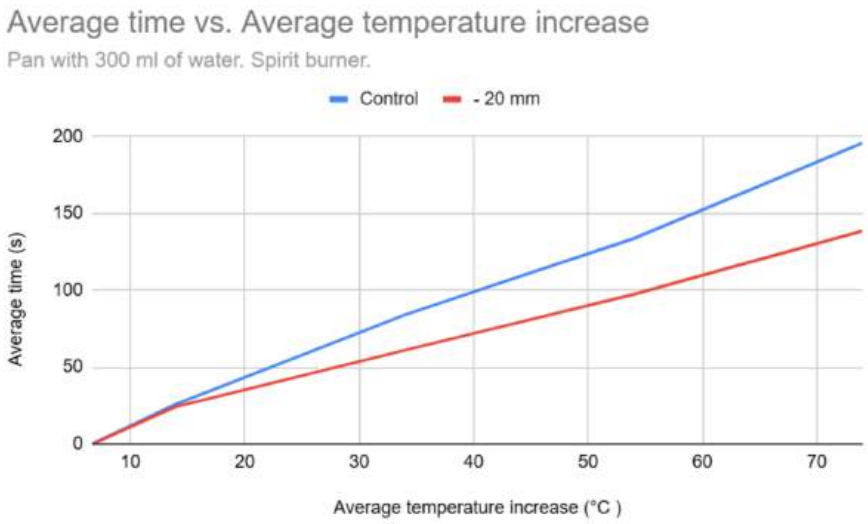


Figure 61: *Results from experiment changing height of pan while heating 300 millilitres of water using spirit burner*

8.8 Discussion on further experiments

At the end of the first round of experiments, it felt that there were several mitigating factors that hadn't been considered. The experiments had not considered how air flow could affect the flame. Gas had also not been used and evaluated against differing heights, along with how a pan height could be improved. Through this collection of experiments these areas have been measured.

The main takeaways from these experiments have been the improvements in heat exchange of the Spirit burner. These improvements however, have not been consistent for the Gas. In the graph in Figure 62 below, the line 0, on the graph, equals the rate of change of heat for a standard position of the pot. The blue and red lines represent the difference in rate of change from the control trials.

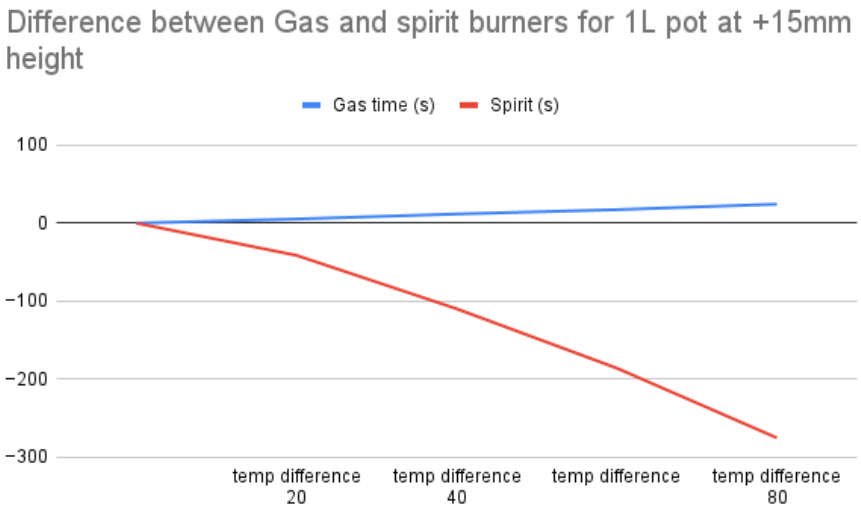


Figure 62: *Graph between Gas and spirit for 1 Litre of water at 15 millimetres*

It can be seen that the gas flame is weaker when the pot is raised by 15 millimetres. This disadvantage however feels nullified when compared to the spirit burner which is far faster at heating the water. For the gas, it takes 24,3 seconds longer for the water to reach 80 degrees. The spirit burner however is 275 seconds faster. This improvement pails the disadvantage to the gas flame.

In Figure 63 the graph can be seen to improve the speed of heat exchange, for both the gas and spirit burner, when heating the pan from 20 millimetres lower than the control.

This improvement is stark, with both the gas and spirit burner improving a vast amount. They improved at 52,7 and 57 seconds faster than the control respectively.

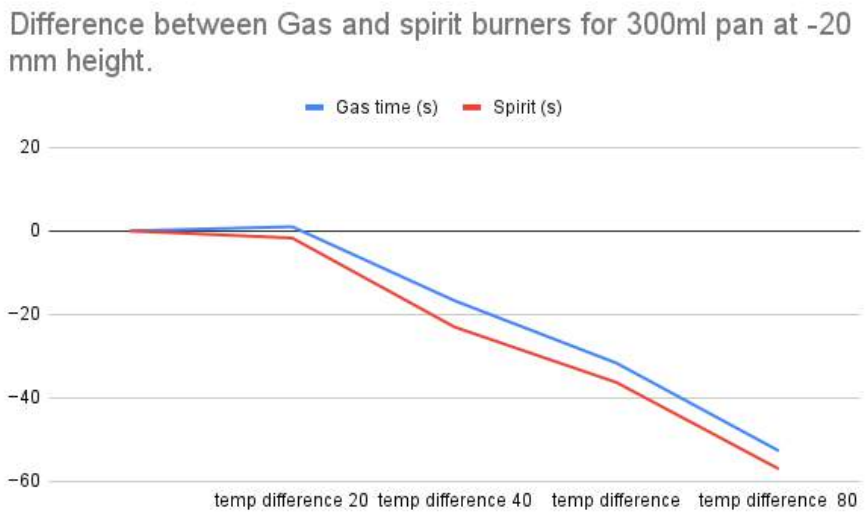


Figure 63: *Graph between Gas and spirit for 300 millilitres of water in pan at -20 millimetres*

To calculate and compare the change from control values, percentage values were found, shown below are the equations used:

$$A = \frac{\text{Control Value} - \text{Height Delta value}}{2} \tag{5}$$

$$B = \text{Control value} - \text{Height Delta value} \tag{6}$$

$$\text{Percentage time change in heat exchange at 80 degrees} = \frac{B}{A} \times 100 \tag{7}$$

Within table below are the percentage changes in length of time for the varying heights for the Pot at 1 Litre and Pan at 300 millilitres of water.

Table 8: Compilation of the results for the second round of height experiments

Experiment	**Improvement in time**
Gas 4/3 Pan -20mm	32 %
Gas 4/3 Pot +15mm	-6,7 %
Spirit Pan -20mm	34 %
Spirit Pot +15mm	41,5 %

9 Brand competition testing

Trangia sent two competition stoves which were probably produced during the 1980's. These stoves used their UWS as the support for housing the pots and pans, rather than any kind of separate CS.

In figure 64 below, the two competition stoves are shown. Both designs have a funnel form where a pot uses its lip to rest on the upper edge of the respective UWS's.



(a) *Front view of competition stoves (Unnamed-Left , Optimus-Right*



(b) *Top view of competition stoves (Unnamed-Left , Optimus-Right*

Figure 64: 1980's competition storm kitchens

To assess the efficiency of these designs, the same 1 Litre experiment was performed so that these stoves could be compared with the original TSK and the enhanced versions with improved heights from the pot. The Unnamed storm kitchen uses a spirit burner similar to the Trangia set, thanks to this, the Trangia spirit burner could be used removing an external factor. The Optimus however, uses a different fuel vessel, soaking up the spirit in a wall of permeable material. In figure 65 below, the tool used to fill the permeable wall is shown. The fuel is poured in and the holes in the tube then allow the fuel to be soaked up by the permeable wall.



Figure 65: *Optimus storm kitchen fuelling tool*

With this difference noted the experiment method did vary between the two competition Storm kitchens. The unnamed Storm kitchen (SK) was first tested. As the distance from the pot to the spirit burner on the Unnamed SK is the same as the +15 millimetres test on the TSK, the results are excellent for comparison.

Listed below is the method run for the experiment:

- **Step 1**: Fill bucket with water
- **Step 2**: Fill second bucket with water
- **Step 3**: Fill and Ignite the spirit burner. After it has been primed, let it burn for three minutes.
- **Step 4**: Fill pot with 1 Litre of water
- **Step 5**: Measure initial temperature of the water
- **Step 6**: Put the pot in the UWS
- **Step 8**: Measure time to reach 20 degrees Celsius
- **Step 8**: Measure time to reach 40 degrees Celsius
- **Step 9**: Measure time to reach 60 degrees Celsius
- **Step 10**: Measure time to reach 80 degrees Celsius
- **Step 11**: Remove the pot from the flame and pour out the water
- **Step 12**: Cool the pot in the second bucket of water
- **Step 14**: Repeat steps 4 through 12 two times

It should also be noted that using this method will lead those that are experimenting to watch how much spirit fuel is left. This experiment can lead to the fuel drying halfway through.

This is the most efficient storm kitchen tested. The unnamed storm kitchen was 62,3 seconds faster to reach 80 degrees than the enhanced TSK. This could likely be down to the pot being held flush with the UWS. This design does show promise for a more efficient storm kitchen. The results from this experiment compared to the 15 millimetres 1 litre test can be seen in figure 66 below.

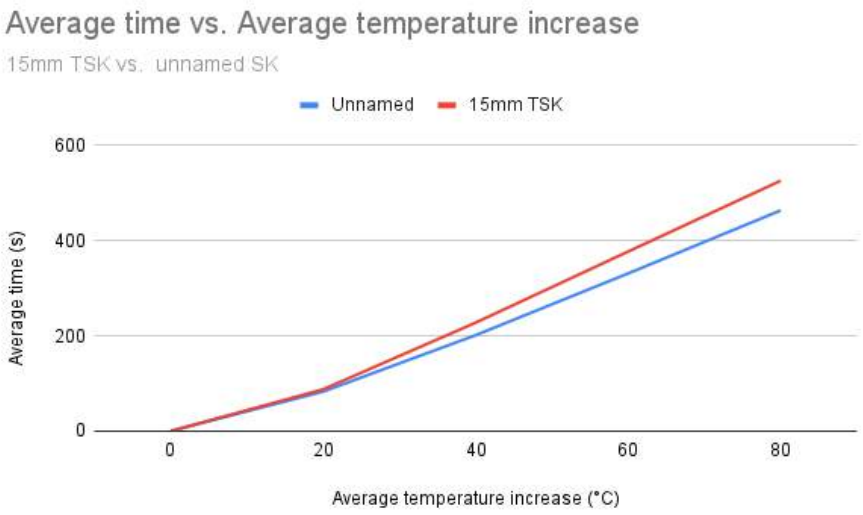


Figure 66: *Results from experiment of Unnamed SK against enhanced TSK with 15 millimetres increased height heating 1 litre of water*

The method for the Optimus storm kitchen is identical to the unnamed storm kitchen except for fuel adding process.

Listed below is the method run for the experiment:

- **Step 1**: Fill bucket with water
- **Step 2**: Fill second bucket with water
- **Step 3**: Fill using the fuelling tool and Ignite the spirit burner. After it has been primed, let it burn for three minutes.
- **Step 4**: Fill pot with 1 litre of water
- **Step 5**: Measure initial temperature of the water
- **Step 6**: Put the pot in the UWS
- **Step 8**: Measure time to reach 20 degrees Celsius
- **Step 8**: Measure time to reach 40 degrees Celsius
- **Step 9**: Measure time to reach 60 degrees Celsius

- **Step 10**: Measure time to reach 80 degrees Celsius
- **Step 11**: Remove the pot from the flame and pour out the water
- **Step 12**: Cool the pot in the second bucket of water
- **Step 14**: Repeat steps 4 through 12 two times

The Optimus was mainly tested for understanding of its function. The heat source is entirely different from the other products and the distance of the flame starts far lower in a concentrated funnel. The product has been compared to the 15 millimetres TSK to understand its relative efficiency. The Optimus was 99,7 seconds slower than the 15 millimetres TSK to reach 80 degrees. This is 162 seconds slower than the other competitor storm kitchen. The results can be seen in figure 67 below.

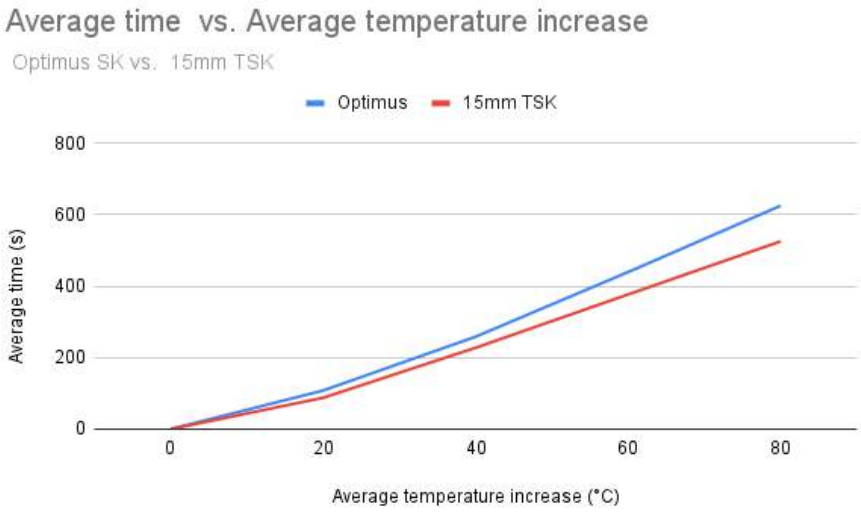


Figure 67: *Results from experiment of Optimus SK against enhanced TSK with 15 millimetres increased height heating 1 litre of water*

10 Specification

Together with Trangia, a requirement specification was developed. It was based on information and input provided from Trangia, as well as knowledge acquired during the research phase. It is shown in Table 9 below.

10.1 Requirement specification

Table 9: Requirement specification for the kettle supports

Number	Statement	Required	Wish
1	The improved solution should be compatible with all previous versions of the storm kitchen.	x	
2	The weight of the new solution should not be higher than the current one.		x
3	The weight should not increase by more than 50 grams.	x	
4	The improved solution should not be more difficult to use than the current one.	x	
5	The manufacturing cost should not be increased.	x	
6	The improved solution should make use of current manufacturing techniques available to Trangia.		x
7	An investment in new machines should not exceed the given budget.	x	
8	The cooking efficiency should be improved.		x
9	The new solution should not contain any loose parts.	x	
10	Reducing production time.	x	
11	Improving manufacturing consistency (tolerances).	x	
12	The container supports must sustain 9,1 kg.	x	
13	The new CS should be congruent with Trangia's brand identity.	x	
14	Prevent the rattling inside the storm kitchen when packed.		x
15	Make the supports compatible with new cooking accessories.		x
16	The improved solution should be more user friendly than the current one.		x
17	The kettle must fit inside the kitchen when packed.	x	

A budget was provided by Trangia based on current manufacturing costs and production time. Requirements 5-7 in Table 9 above, as well as 10 and 11, are based on this information.

11 Ideation

This chapter presents the development of the design through the previous shown research. This involves sketching and concept development plus manufacturing specific evaluation.

11.1 Initial Sketching

The sketching phase was used to formulate solutions, allowing for broad creativity. This led to new ideas which could be argued would not fully function for the use case but developed new avenues. Below are two of the key avenues presented through rough early sketches.

In Figure 68 below, the collection of images reflect the idea of finding the simplest solution for the design brief and specification.

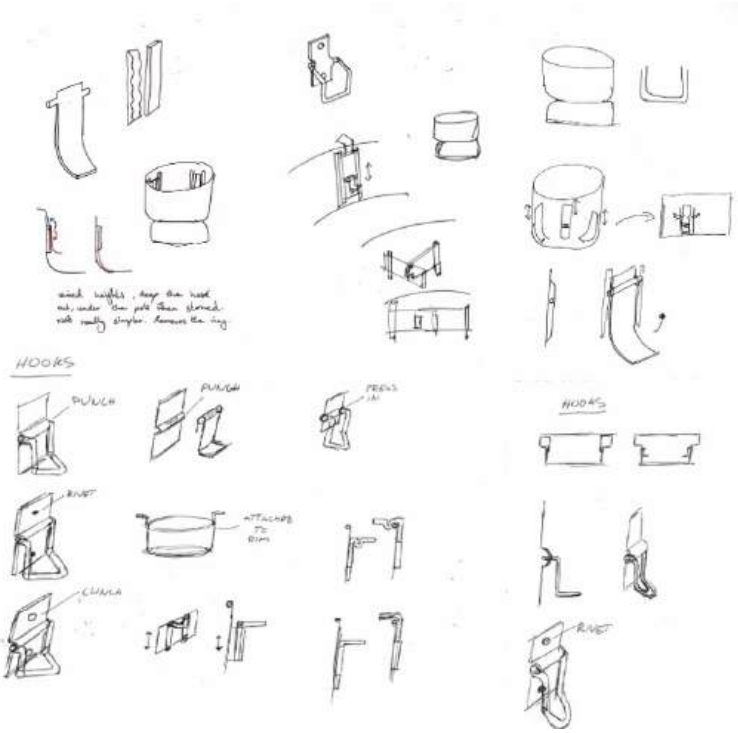


Figure 68: *A collection of sketches of hooks and securing methods*

It can be argued that the easiest solution for the tolerance issues and manufacturing difficulties for the CS and ring are through a similar hook method. In the sketches shown, it is proposed that Trangia use a fixing method to hold a hook. This fixing

method could involve using several of the existing processes: clinching, riveting, clamping and press stamp-cutting.

For the other avenue, it has been argued that a way to remove the need for a vertically rotating hook would remove several issues. With this in mind a solution using a T form was proposed. This overall form and design elements can be seen in Figure 69

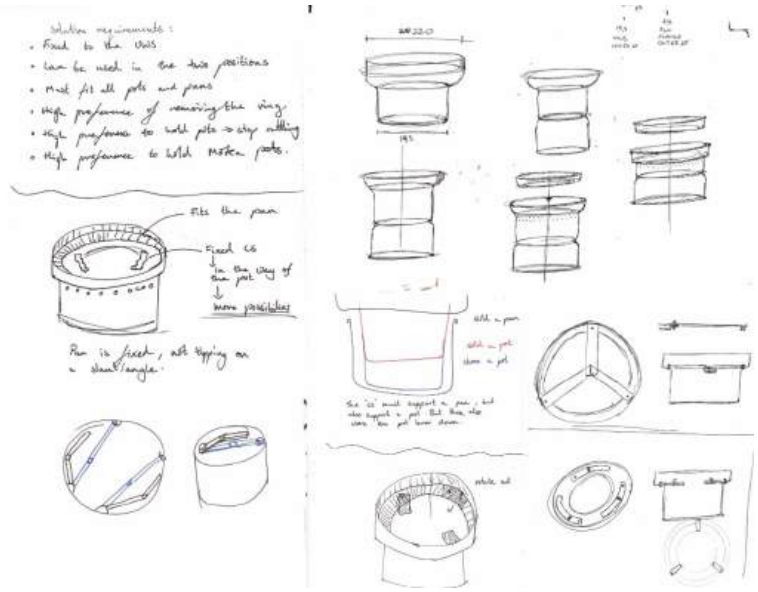


Figure 69: *A collection of sketches of the T form*

The T shape of the UWS would naturally have a lip which the pan could then rest on for cooking. This removes the need for a hook to rotate upward. A hook can then be rotated horizontally and be kept more easily out of the way.

11.2 PURSE Method

To diverge in the ideation phase a tailor made method was formulated. This method takes aspects from the crazy 8 method, decision matrices and QFD.

PURSE is an anagram comprised from it's five elements:

- Problems- Issues with the current design
- Uses- What are the design's functions
- Requirements- Needs from the design
- Solutions- Possible design ideas
- Evaluation- Which Ideas should be pursued further

Within Figure 70 below is the filled in form for the Purse method up until the solutions stage. For the solutions stage, the requirements are numbered and then three of the requirements are randomly combined as a prompt for a sketch. All

three requirements must be met but any other needs can be ignored. This stage is repeated five times, with 5 minutes per sketch. This totals as a pair, for 10 new design solutions.



Figure 70: *PURSE method form filled in*

After this stage the solutions can then be evaluated and considered for further development. One design was found to be promising. This Sketch can be seen in the upper half of Figure 71 .

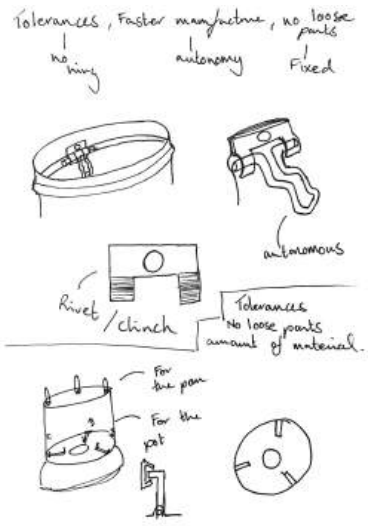


Figure 71: *Example sketch from the solutions/crazy 5 stage*

This hook solution, intends on solving the requirements: Better tolerances, Faster

manufacture and no loose parts. The proposal uses the current UWS, including the inner ridge.The hook is made from autonomously bent wire, a stamped attachment and uses either Riveting or clinching to secure to the UWS.

11.3 Concept refinement and convergence

Within this section, potential concepts are presented and then evaluated. The designs from the Purse method are refined and then evaluated using a Pugh matrix.

Listed below are the compiled list of design concepts:

The T- Form is a refined version of the design found in figure 69 , the concept is also explained within chapter 11. The refined drawings in figure 72 below present the silhouette and proportions to the pan.



Figure 72: *Concept sketch of the T-form*

A 3D model was created to confirm that the form would function as intended. Images of the model can be seen below in Figure 73.



(a) *T-form parts* (b) *T-form packed* (c) *T-form pot*

Figure 73: Images of the T-form 3D model

The T-form functions well, with the pan sitting within the UWS. One concern is the increased exposure of the pot. This could increase the effect of the wind. The 3D print has however proven that the geometry of the design is possible.

The recessed hook concept can be seen in figure 74 below, the concept uses the current UWS and requires minimal manufacture changes. The design has also been described in the previous chapter, 11.2 and figure 71.

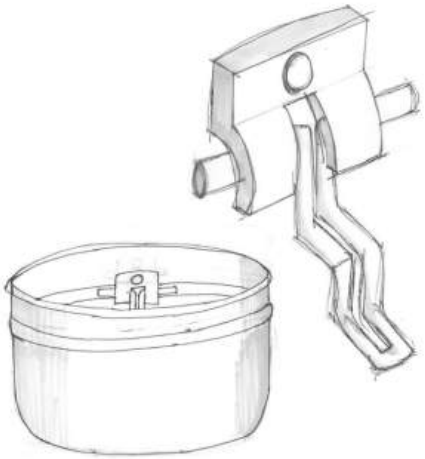


Figure 74: *Concept sketch of the Recessed hook*

Sliding platform is a novel approach for the CS. The design uses two platforms, connected with a hinge. The two parts are housed in rails. The design gives a range of heights to maximise the flame efficiency. A drawing of the concept can be seen in figure 75 below.

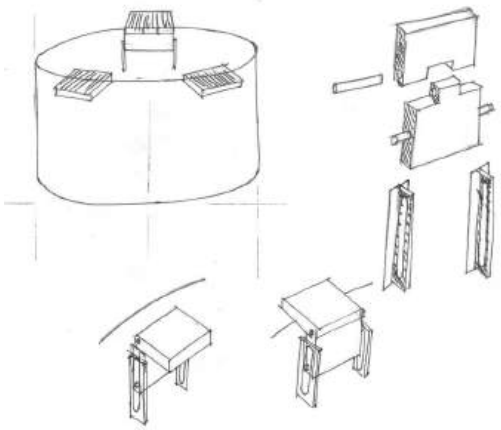


Figure 75: *Concept sketch of the Sliding platform*

Similar to the sliding platform, the fourth concept, Sliding wire allows varying heights to support the pots and pans. A wire is autonomously bent into an L form. This wire is then housed between the UWS and a stamped metal sheet secured by rivets. The wire can then be pulled to a height and rotated to orientate for a pan or pot. A sketch of this concept can be seen in figure 76 below.

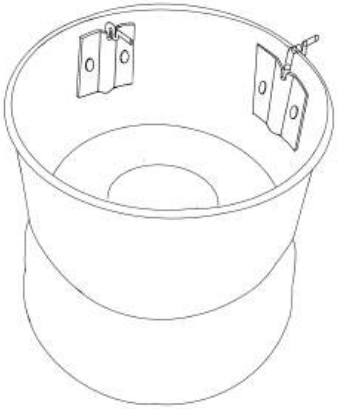


Figure 76: *Concept sketch of the Sliding wire*

Rivet hook can be deemed the simplest solution. The concept uses an autonomously bent wire which has the same side form to the current CS. The wire is then housed in a humped stamped sheet, riveted to the inner of the UWS. This concept can be seen in figure 77 below.

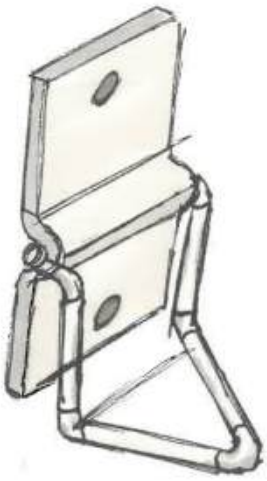


Figure 77: *Concept sketch of the Rivet Hook*

Stamp cut harnesses a very different manufacturing technique used by Trangia. The design requires stamp cuts made into the UWS creating horizontal channels on the

inner of the UWS. The channels then fix autonomously bent wires like the Rivet hook and Recessed hook. A sketch of the design is presented in figure 78 below.

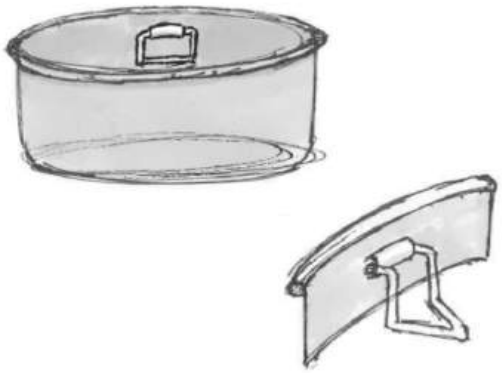


Figure 78: *Concept sketch of the Stamp cut*

11.4 Evaluation-Pugh matrix

In figure 79 below, a table using the Pugh matrix is presented. This method helps to find the most appropriate concepts based on relevant criteria. The original design is used as the baseline and all concepts are rated against it to find the best alternative with the highest positive number [25].

Concept	Manufacture speed	Ease of use	Height variation	Longevity	No. of parts	Manufacture convert	Material & weight	tolerance	Brand	Added value	Overall score
T-Form	2	1	2	1	1	-2	3	2	0	3	13
Recessed hook	2	0	1	0	1	-1	2	3	0	1	9
Sliding Platform	-3	-2	1	-1	-3	-3	-2	-2	1	3	-11
Sliding wire	1	0	1	0	1	-1	3	1	0	1	7
Rivet Hook	2	0	1	0	1	-1	1	2	0	1	7
Stamp cut	3	0	1	-1	2	-	3	2	-1	1	8

Figure 79: *Pugh matrix for the proposed concepts*

The Pugh matrix showed that certain concepts should be continued and others dropped at this stage. This was particularly pertinent in regards to the T-form and sliding platform. The T-form had a very positive score, whereas the sliding platform very negative.

11.5 Feedback from Trangia

The concepts that were evaluated in the Pugh matrix above, minus the Sliding platform, see Figure 79, were sent to Trangia for review of their manufacturability and functionality.

Below is presented the feedback for each concept.

T-shape

- The T-shape required a big change to the current design of the UWS. For this to be a viable solution, there would have to be clear functional advantages for the consumer.
- The increased wind protection for the pan is a clear advantage.
- Solutions for the container supports have to be developed in order to evaluate the concept further.

Recessed hook

- There was a risk that the solution felt cheap, mainly because the wire looked and felt weak.
- Further development of the concept where other manufacturing methods are considered is necessary. For example, stamping the support rather than making it from wire.

Rivet hook

- Much like the recessed hook, there was a concern that the solution might look a little cheap due to the wire.
- Trangia wanted to see if there was a way to make the solution feel more "premium".
- Further development of the concept was necessary.

Slidey wire

- The concept had an advantage over the current one due to its design stopping it from falling down.
- The manufacturing method was slightly unclear.
- There was a fear of the support rattling when the TSK was packed.
- Further development was necessary, focusing on manufacturing and potential rattling.

Stamp cut

- The main concern for this concept was how to make the outside of the UWS look good with the cutouts.

- A decision was made to leave this concept due to the necessary changes to the UWS.

In addition to the feedback on the concepts, Trangia did an experiment of their own, varying pot heights with the spirit burner. During this, they came to the conclusion that an increased height greatly increases the amount of soot on underside of the the pot. This was something that had not been experienced during previous experiments, most likely due to the use of a different fuel, see figure 80 for the used fuel. The results from the experiment are shown in Figure 81 below. The second pot shows the trial with a raised pot, compared to the first pot which shows the current height. Pots three and four show the effects of raising the pot with a small portion of water added to the fuel, which is common practice for minimising soot levels.



Figure 80: *Red spirit used by the Trangia team*



Figure 81: *Underside of pots after experiments varying the height*

After some discussion it was apparent that to get a conclusive result, every type of fuel on the market had to be tested. This would result in additional experiments that would take up a significant amount of time. Due to this, it was decided that no further experiments would be conducted, and the focus would be placed on improving the CS.

11.6 Reflections on concepts

Upon reviewing the concepts, it was realised that some of them would not work with their current design. The key issue has been the attachment to the UWS, in most of the concepts the hook is obstructed when rotated upwards. Figure 74 is a perfect example of this. The design however has been iterated and now functional as can be seen in figure 82 below.

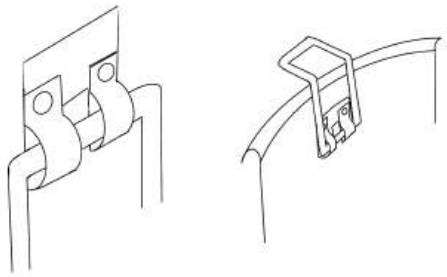


Figure 82: *Iterated recessed hook*

These issues have often arrived and it has been deemed that 3D modelling or physical prototypes help present the issues quicker. Additionally it was realised that the concept 'Slidey wire' would not function due to its required length. The length would need to be over 100 millimetres to function for both heights. An alternative to this design was iterated and can be seen in the following chapter.

11.7 Second round of ideation

The following concepts are based on Trangia's insights and reflections, plus issues found, mentioned in the previous chapter.

Original Hook

The original hook design uses the stamped hook from the current TSK but uses a double riveted attachment rather than the ring used today. The proposed design can be seen in figure 83. Listed below are the positives and negatives of this design:

- Use old stamped hook (no need to change tooling) .
- Easy to attach (two rivets)
- familiar design for users
- Less material
- Might be difficult to bend the hook around the attachment
- No improvement for users



Figure 83: *Original UWS, original stamped hook*

180 Snake

The 180 snake design, is the developed version of slidey wire. Using a autonomous bent wire maneuvered 90 degrees and fixed on the lip of the UWS. The design can be seen below in figure 84. Listed below are the positives and negatives of this design:

- Improvement for user (lowering pan, hook cannot fall down during use)
- Easy to attach (two rivets)
- Automated manufacturing (wire bending, stamping)
- Might look weird and flimsy
- Unfamiliar for users

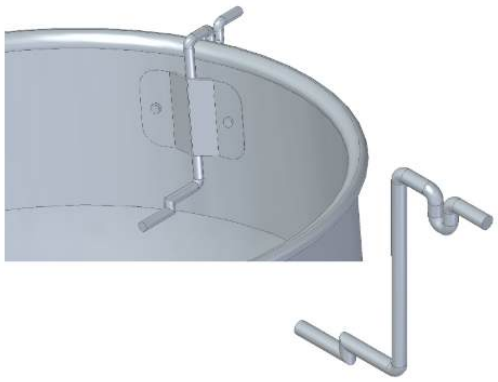


Figure 84: *original UWS, 180 degree curved wire support*

180 Flag

180 flag is a more solid version of 180 snake. It is likely more sturdy but possibly more expensive to manufacture. The design can be seen below in figure 85. Listed below are the positives and negatives of this design:

- Improvement for user (lowering pan, hook cannot fall down during use)
- Easy to attach (two rivets)
- More solid than the 180 snake
- No need to change position of the CS when using pot or pan
- Might look weird and flimsy
- Both parts require complex manufacture

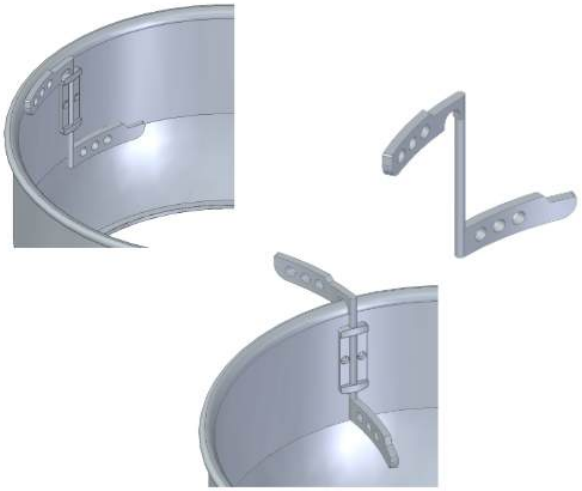


Figure 85: *original UWS, 180 degree curved solid support*

Hooked Seat

The hooked seat is a more sturdy version for the current stamped hook. The attachment will not require as tight tolerances as the current design, plus appears more robust. The design can be seen below in figure 86. Listed below are the positives and negatives of this design:

- Solid part, looks robust
- Easy to attach
- Familiar function to users
- Pan height can be lowered
- The hook likely has the same old issue (flopping back down)
- Combining a solid part with wiring (cast)

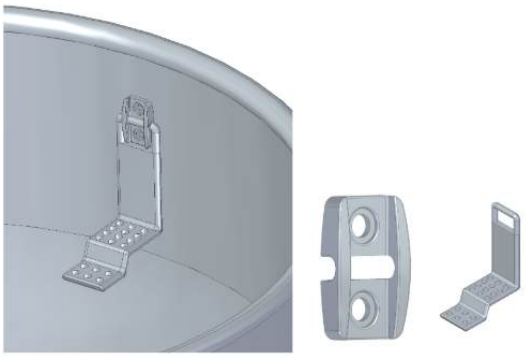


Figure 86: *Original UWS, Seat with wire attachment*

Bent Wire double Rivet

The bent wire double rivet (BWDR) uses an autonomously bent wire and double rivet attachment. The design can be seen in figure 87. Listed below are the positives and negatives of this design:

- Automated manufacture
- Familiar function to users
- Easy to attach (two rivets)
- height of the pan could be changed
- The wire might look flimsy
- Could flip down



Figure 87: *Original UWS, double riveted hook*

Bent Wire Single Rivet

The bent wire single rivet (BWSR) is similar to the previous design (BWDR). The attachment however is connected with one rivet with two wings prevenetiing the rivet from rotating. The design can be seen below in figure 88. Listed below are the positives and negatives of this design:

- Easiest to attach (1 Rivet)
- Familiar function to users
- Do not know if this is actually possible
- Could flop down
- Might look flimsy



Figure 88: *Original UWS, single riveted hook*

H-arrow

The H-arrow design uses two cast parts, one being an arrow shape, the other the attachment. This solution is only applicable in combination with the T-form. The design can be seen in figure 89 below. Listed below are the positives and negatives of this design:

- Looks cool
- Robust
- Easy to use
- Might rattle (can be solved by redesigning)
- Has to be cast (at least with current design)



Figure 89: *T-form, horizontal arrow hook*

Carabiner

The Carabiner design is inspired by a Carabiner spring lock. The two wires are under tension and will natural flip at 90 degree points. This creates a sturdy and satisfying hook. The design can be seen below, in figure 90. This solution is only suitable in addition with the T-form UWS. Listed below are the positives and negatives of this design:

- Looks cool
- Automated manufacturing (wire)
- The supports snap into place
- Might look flimsy
- Cast attachment
- Assembly might be tricky



Figure 90: *T-form, horizontal carabiner hook*

11.7.1 Feedback from Trangia

The concepts presented in section 10.7 above were sent to Trangia for review. After reviewing, a meeting was held with Trangia where the concept were discussed further.

Attachment

Many of the proposed concepts would be using rivets as the main attachment method, something that Trangia were critical to. It was also stated that the current ring used for attachment might not be completely out of the picture, as previously thought. It became apparent that the issues with the ring did not lie in the principle itself. It was rather a matter of how the current CS are attached to the ring that created the need for such tight tolerances. Consequently, another round of ideation was proposed, with a focus on different attachment methods.

T-form

Upon discussion, it was further decided that the T-form was no longer a viable solution. The change in the product's visual expression would simply stray too far from the traditional silhouette. In addition, it would require an extensive alteration of production. Therefore, no further development efforts will be made in the T-form.

Container supports

Of the various types of CS presented to Trangia, the "180 flag" was deemed the best (see Figure 85 above). It was agreed that this solution would improve the user experience, as well as the durability of the product. It was therefore agreed that the "180 Flag" was the preferred solution, and refinement of the concept would be the next step.

Conclusion

During the discussions it was concluded that another round of ideation would take place, focusing on different attachment methods, in combination with the "180 Flag".

11.8 Third round of ideation

The following section presents the third round of ideation, focusing on the attachment method of the "180 Flag".

Flat Ring

The Flat Ring uses the current groove in the UWS for attachment. A strip of metal is stamped and rolled into the correct shape, and the CS are held behind the ring. The concept is shown in Figure 91 below.



Figure 91: *A 3D-printed prototype of the Flat Ring*

The model was used to confirm the function of the Flat Ring, together with the 180 Flag. It was concluded that the concept worked, but required further iterations of both ring and flag.

Clamp and Channel

The Clamp and Channel uses three holders clamped around the edge of the UWS, and a ring in the current groove. The clamping ensures that the holders stay in position, and the ring holds them firmly against the wall of the wind shield. In order to maintain a low enough profile of the attachment, the ring likely has to be made from a thinner wire. However, this is something that will be examined during further development. The concept is presented in Figure 92 below.

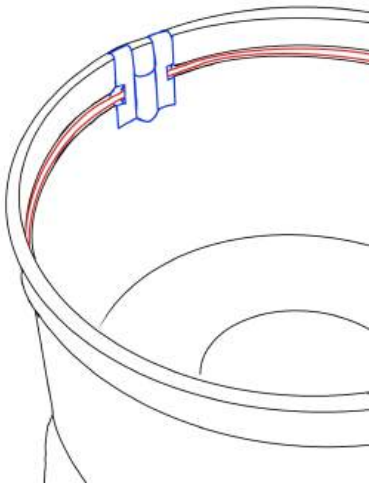


Figure 92: *The Clamp and Channel concept on an UWS*

Clamp and Rivet

The Clamp and Rivet is similar in construction to Clamp and Channel. The difference is that a rivet is used instead of the ring to hold it to the inside of the UWS. This results in removing the current groove in the UWS, which will alter the silhouette of the TSK. The Clamp and Rivet is shown in Figure 93.

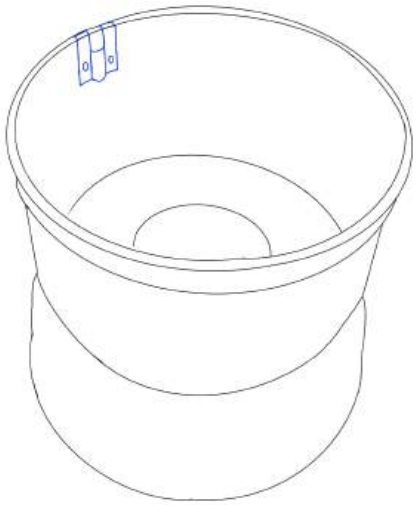


Figure 93: *The Clamp and Rivet concept on an UWS*

Stamp Cut

The Stamp cut takes the stamp cut process used on the LWS for the strap holders. The Stamp cut uses material from the UWS pushing a semicircular form externally. Holes are then cut, creating a channel to hold the 180 Flag. This attachment will add no weight and be manufactured completely autonomously. The design could however appear rough and unattractive. Although the design element does have precedent on the TSK. The design can be seen in Figure 94

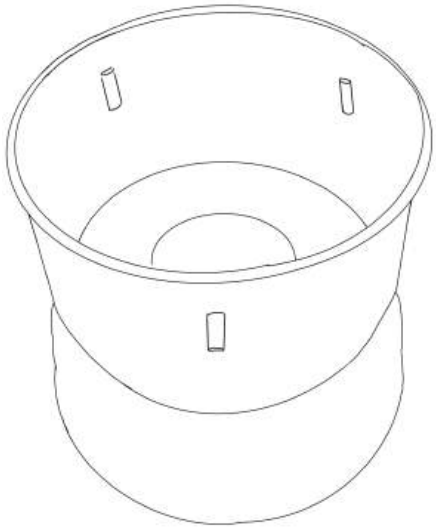


Figure 94: *The Stamp Cut concept on an UWS*

Double Channel

The double channel uses the current UWS and the groove currently used for the retaining ring. The double channel is secured using a smaller ring, which retains stamped plated through the horizontal channel. The vertical split channel then holds the 180 Flag. The double channel can be seen in Figure **??** below.

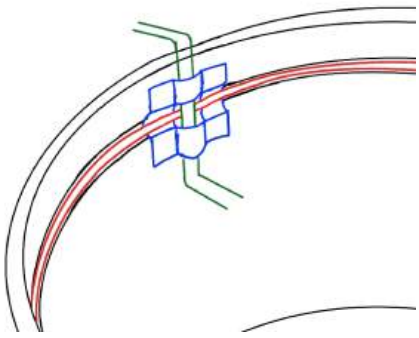


Figure 95: *The Double Channel concept on an UWS*

11.8.1 Feedback from Trangia

Yet again the developed concepts were sent to Trangia for evaluation and feedback. Mainly because of its simplicity, it was concluded that the "Flat Ring" concept (shown in 91) was the strongest solution. In a meeting with Trangia, it was decided that the final concept would consist of the "180 Flag", in combination with the "Flat Ring". This combination can also be seen in Figure 91 above.

It was also recommended that the manufacture of the hook and ring should be made as simple as possible. Preferably with the use of Trangia's current stamping press.

12 Refinement

The following chapter details the finalisation of the design for the new model container supports. This includes: Sketched designs, CAD modelling, DFM, FEA, DFA and cost analysis.

12.1 Flat Ring development

The development of the Flat Ring was done through several iterations, each optimizing mainly the weight and function. Many of the designs were 3D-printed to better visualize them. The main design parameters are shown in Figure 96 below. These will be used to better understand the development of the component.

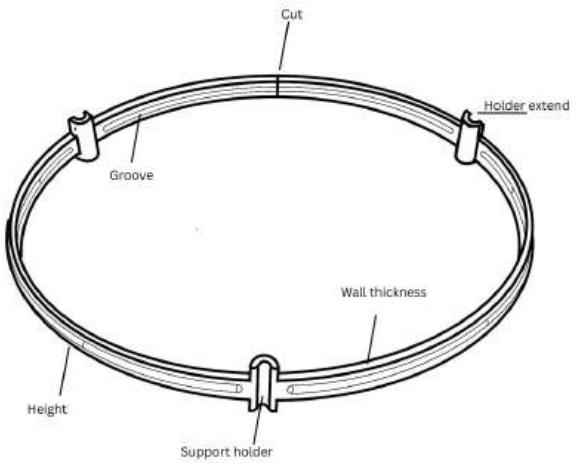


Figure 96: *The geometrical parameters changed during development*

Below is a summary of the development of the Flat Ring, following the version history.

Stage One

The first stage of the development was mainly used as a proof of concept. A model was created using CAD, and a 3D-printed version was produced. This was placed in the UWS to test the function of the 180 Flag, together with the Flat Ring. This can be seen in 97 below.



Figure 97: *First iteration of the flat ring*

The printed model was two millimetres thick to ensure a sturdy print, a value which later had to be reduced to lower its weight. The function of the concept was deemed satisfactory, and the development continued to the next stage.

Stage Two

The second stage focused on reducing the weight of the part. The height was decreased and the thickness was reduced. It was realized that the extensions for the flag attachment points might not have been necessary, and subsequently the were removed. To test the changes, another prototype was made using the 3D-printer. This version can be seen in Figure 98 below.



Figure 98: *Second iteration of the flat ring*

After testing the ring in an UWS, it was concluded that the extensions were superfluous. The ring itself offered enough sideways support for the 180 Flag, and the extensions were permanently removed.

Stage Three

To confirm that a flat ring was the optimum shape, a wire ring was 3D-printed for functional testing. Like the other versions, this was placed in an UWS together with the 180 Flag. The result is shown in Figure 99 below.



Figure 99: *Third iteration of the ring using a wire*

As expected, the wire ring did not provide enough support for the 180 Flag, and it tilted sideways as it was moved vertically. This design was therefore discarded.

Stage Four The fourth version had further reductions of the material thickness to lower the weight. The aim was to match the weight of the current ring used by Trangia. A thinner version was printed to understand how it would look. This is shown in Figure 100 below.



Figure 100: *Fourth iteration of the flat ring*

The thinner ring was considered feasible, but it was also realized that it would be impossible to decide a thickness without making a working prototype.

Final stage

Before the final stage, the concept was discussed with the product developers at Trangia. It was realized that the shape of the groove in the UWS could be changed to fit the entire flat ring. This meant that the groove in the ring could be removed, which would be advantageous in the manufacturing process.

It was also discussed how to decide on the material thickness. Trangia had suitable material in store, which was sent over to use for prototyping. The final version of the ring was therefore based on this material. The result can be seen in Figure 101 below.



Figure 101: *Final iteration of the flat ring*

12.2 180 Flag- Profile

At this stage the 180 flag function has been defined. Although the profile of the component requires definition. In Figure 102 below the confirmed parts of the form is shown. This outline was used as the starting point for the sketch proposals.

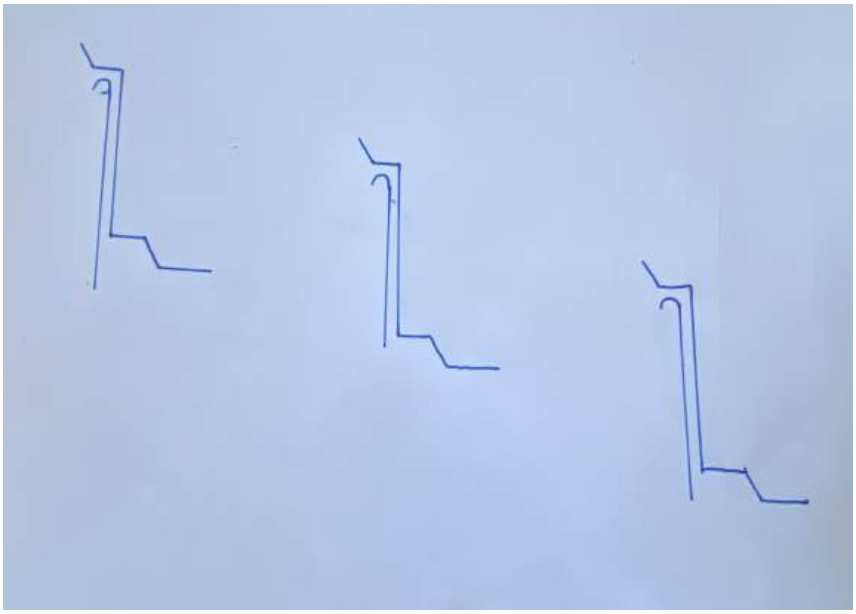


Figure 102: *Confirmed part-outline for the 180 flag*

The initial sketches from the outline and further work can be seen in Figure 103 below.

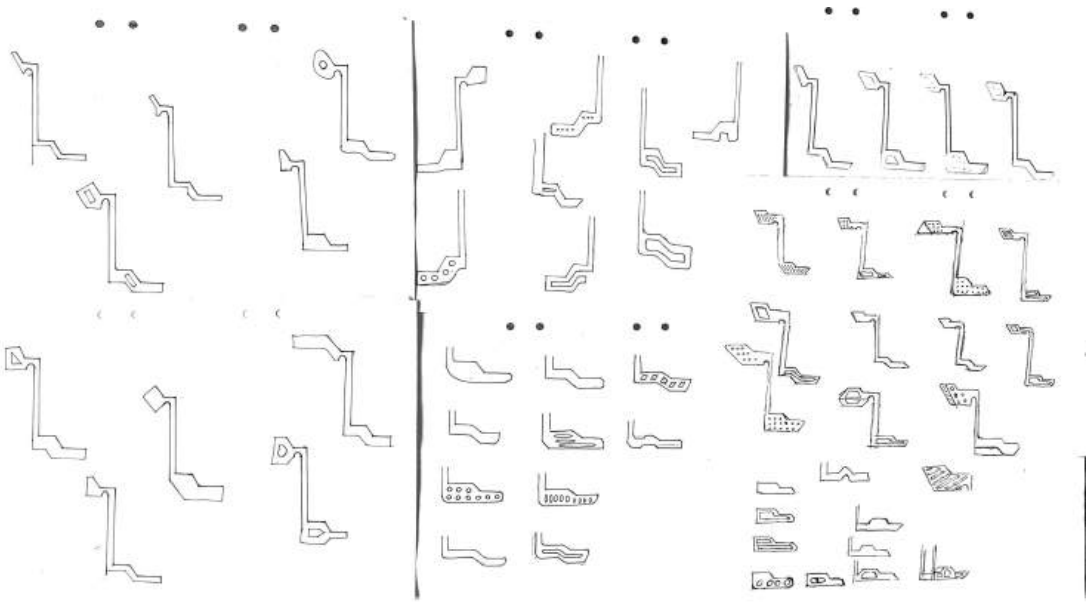


Figure 103: *Possible sketches for the 180 flag*

These sketches were reviewed against the descriptions defined in section Design Expressions 5.2. Based off these definitions it was determined that the design shown in Figure 104 below was most appropriate.

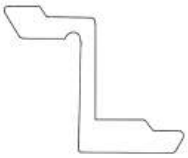


Figure 104: *The 180 Flag chosen profile*

This profile for the 180 flag aligns well with the keys values of the company: Simplicity, Quality and Reliability. The design has a form which follows the function ensuring simplicity. Quality is assured with a form that is robust and lacks thrills.

Finally, with solid, smoothly edged shapes, the form is dependable, a hallmark of reliability.

The profile was proven to also function for all of the components reliant on the CS. This is shown in Figure 105 below.



Figure 105: *Test of the profile on the containers*

12.3 180 flag - Weight reduction designs

In this section the process of designing in the weight reduction elements of the 180 flag is shown. The aim being for the CS to not exceed 50 grams, compared to the original design. Using the outline of the 180 flag profile, see Figure 104, proposed sketches were created. The Sketches were taken as three different styles to maximise variation and extend the possible options.

Style 1 - Angular and offset

The first style uses the offset of the profile and triangular forms to maximise the weight reduction. The designs can be seen in Figure 106 below. They can be considered very angular. This possibly would appear too close to Primus' aesthetics.

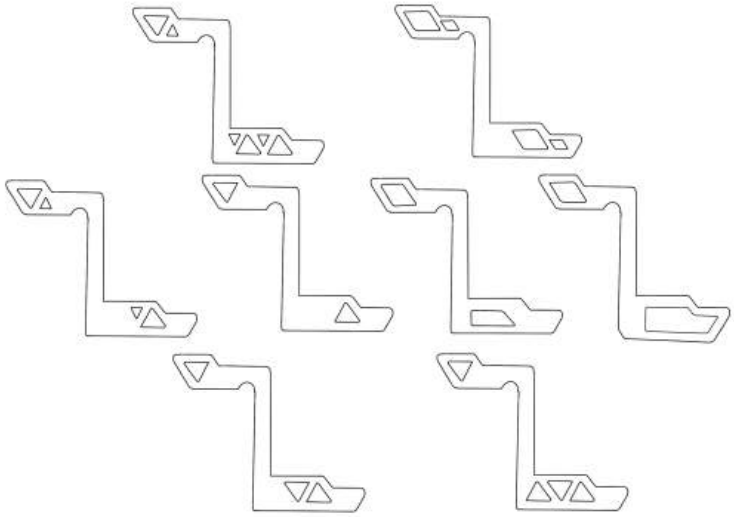


Figure 106: *Angular and offset sketches*

Style 2 - Slot

For the second style, the weight of the 180 flag is reduced through patterned slots. The design an be seen in Figure 107 below. The style, however, didn't feel similar to Trangia's current product offering. The slots implied fast performance similar to a sports car or high performance laptop.

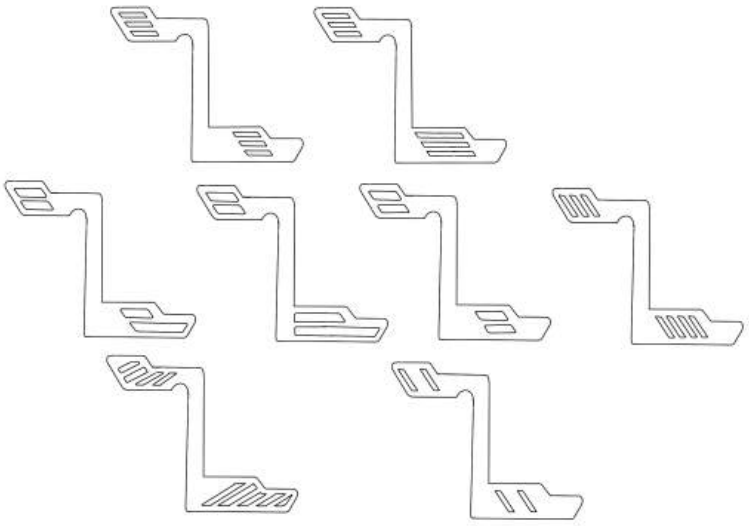


Figure 107: *Slot form sketches*

Style 3 - Circular The Circular style as shown in Figure 108 below uses repeated circles. This felt far more appropriate for any brand change for Trangia. The design is similar to the Tamoj which can be seen previously in Figure 27 .

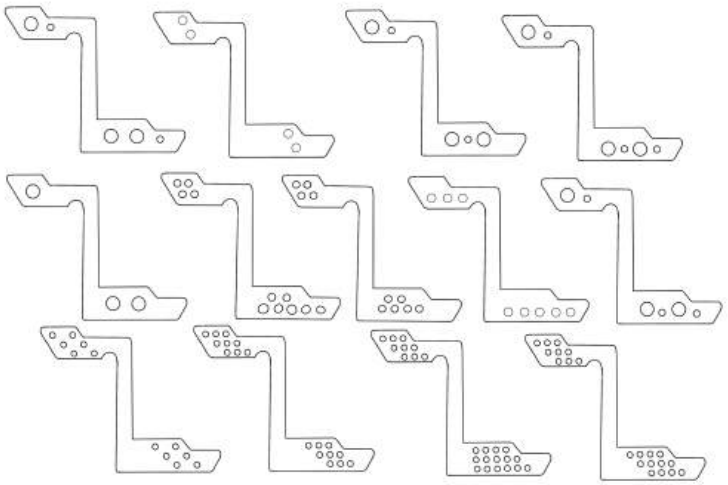


Figure 108: *Circular form sketches*

For the final design, an option was taken from the collection of sketches in Figure 108. The design felt crowded in a 3D sketch. This led to reducing the bottom cut outs to one 7 millimetre hole and one 3.5 millimetre hole tangential from each other. The sketch of the design is shown in Figure 109 below.

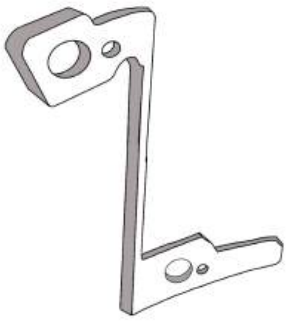


Figure 109: *Final sketch of the Flag 180 form*

12.4 180 Flag - Function surfaces

To imply the function for a product it is possible to use functional surfaces to imply the use for a component. In 'A Short Course in Industrial Design' it is recommended to use surface variation to emphasize the functional surfaces of a part [26].

Presented in Figure 110 below are images from the CAD of the proposed functional surfaces. These surfaces are on the Bottom and top touch point, and are expected to be created during the die cutting manufacturing process. This means that the functional surface must be created along the whole surface, like channels, and not like knurling.



Figure 110: *Function surface options. Left to Right: Doris, Sigge, Gösta, Per, Steffe*

It was determined that Doris, was most appropriate for Trangia's design expression and identity. The functional surface is 0.5 millimetre circles spaced apart with a 0.1 millimetre fillet on all edges. This relates well with Trangia's sturdy and simplistic design style.

12.5 180 Flag - 3D print development

The section below, details the development and iterations of the 180 flag based off the design presented previously, which can be seen in Figure 91 in the section above.

To better understand the development of the 180 flag, Figure 111 below states the name for all the features of the design. These feature names will be used during the explanation of how the design has been iterated.

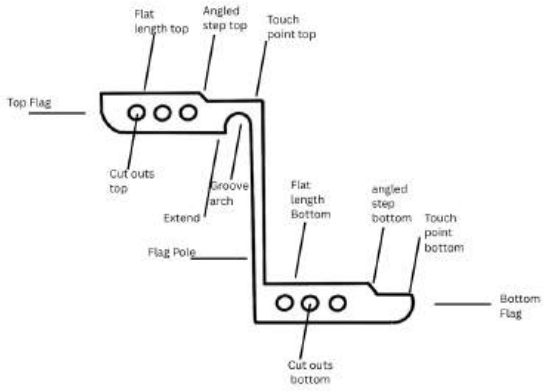


Figure 111: *Descriptive image of the 180 Flag*

In Figure 112 below, all of the stages of the hook can be seen. Each stage presents several iterations. The 3D prints were used to assess the perfect dimensions for the pot and pan. They also were used to assess the aesthetics of sketch proposals, plus thickness of the model.

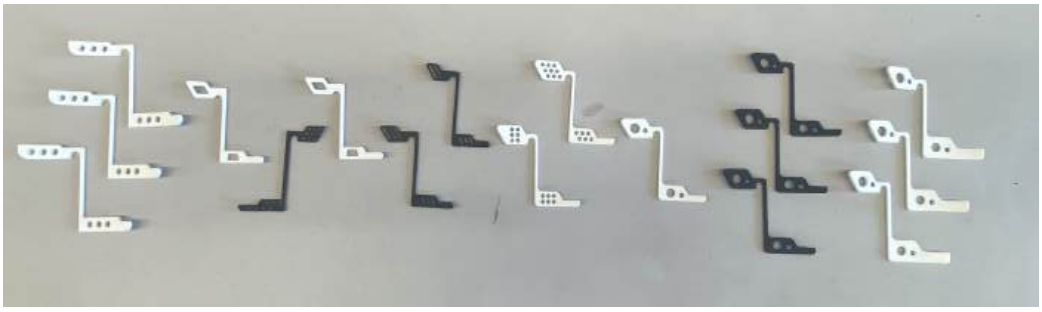


Figure 112: *All Eleven 3D printed stages of the 180 flag development*

Stage zero

Stage zero was a 3D print of the first proposed form of a hook with this function. The wire form is the basis for the profile used for future stages. The design however could appear flimsy and led to the led to the need for a flag body form.

Stage One

Stage one was the first iteration of the 180 flag, a development from the wire design. The design was used to confirm the length of the flag pole, and the touch points, against the existing solution. It continued using a 3 millimetres diameter for the flag pole.

Additionally Stage One considered the use of cut outs. However, this proposal was created before any sketches and considered as a purely functional choice.

Stage Two

It was noted that Stage One flags could be made far shorter. Stage Two is the first iteration with the overall form that would be used for all future stages. The angled steps are parallel at 120 degrees and at a height of 4 millimetre. The Arch has been reduced to a diameter 0.1 millimetre larger than the UWS ridge. Finally the touch points lengths have been reduced aiming for a tighter hold on the pot and pan.

Stage Three

Stage three differs from the rest of the stages as it mainly considers the practicality of a 1 millimetre pressed version. The design does however also increase the depth of the arch. This creates a lip and reduces how easily the 180 flag can ride up when rotated.

The design also trials angled holes, this was performed to test a hole design proposed as shown in the previous section.

Thee 1 millimetre pressed version, uses a u-shaped flag pole. The whole body is pressed from one sheet of metal. The design functioned, but it was considered flimsy and non-congruent with Trangia's values.

Stage Four

Considering that Stage Three was flimsy, a 2 millimetres print was trialled instead. It was also realised that a flat form where the flag pole is square should be trialled.

Stage Five

As Stage fours square flag pole functioned well, the following Stages also used a square flag pole. The 180 flag flips between the two positions, making the design feel solid and well considered. A further iteration at Stage Five were the touch points incrementally closer to perfectly fitting the pan and pot. This process required many iterations as any dimension change was multiplied by affecting all three 180 flags.

Stage Five also tested the use of slots rather than following the profile or holes for cutouts

Stage Six

Stage six trialled the design for a 27 TSK. The design requires a smaller diameter and strong angle about the flag pole. The iteration was however imperfect, with distances of the touch points differing from Stage Five

Stage Seven

Stage Seven increases the height between the touch point and Flag length for the top and bottom. The pan and Pot were able to ride upwards on the 4 millimetre height difference. The difference was increased to 7 millimetre. This increased height also allowed for more cutouts, leading to trialling a 9 hole design on the top flag.

Stage Eight

Stage Eight was used as an aesthetic comparison of Stage seven. It had the same profile as Stage Seven. As any dimensional change required three 180 flags, this iteration was considered advantageous.

Stage Nine

Stage Nine presents the finalised cutouts for the 180 flag. It also increased the length of the arch depth from 1.5 millimetre to 2.5 millimetre. This ensured that the top flag didn't rotate and ride upwards when in use.

Stage Ten

Stage 10 is the finalised version of the model for the touch point lengths. The pot and pan sit perfectly on the touch points and against the angled steps.

Stage Eleven

In Stage 11 the functional surfaces are added, used to signify where the user should place the containers.

Final stage

During the final stage, three 180 flags were made in steel using a water jet. The steel was not stainless, and had unknown strength properties, but could be used to test how the supports would feel in metal. The final version is shown in Figure 113 below.



Figure 113: *Metal version of the 180 Flag assembled on an UWS*

During testing with the metal version, it was concluded that the 180 flag would wiggle in its cooking position. This posed no real issue for the function, but it made the solution feel insecure. However, after further examination it was noticed that the supports were more solid in certain positions around the rim of the UWS. After discussions with Trangia, it was concluded that this was due to manufacturing inconsistency of the UWS. Since the edge of the UWS had not been important for function before, its manufacturing tolerances were relatively inexact. For the new solution to work, this would have to be addressed in the future. Additionally, it was observed that the waterjet had some inconsistencies during manufacturing, and each support was slightly different. These fitment issues would have to be addressed during further refinement of the product, and its manufacturing process.

12.6 180 flag - Manufacturing methods

For the 180 flag the primary aim was to reduce the manufacturing time through autonomous methods and a limited number of steps. An additional aim was to manufacture using existing machines in the Trangia factory. The available manufacturing methods can be found in a previous chapter:7.1. Finally the design would also need to adhere to the specification requirements. This primarily included No.3 : 'The weight should not increase by more than 50 grams and No.12 : 'The container supports must sustain 9,1Kg within Table 9.

Listed below are the proposed manufacturing methods for the 180 flag. All the proposed methods would require tumbling as an additional process to soften the edges of the part.

Initial proposal

The first proposed design was created prior to the use of the requirements detailed above. The design is created with the aim that the 180 flag is made as a cast component. The design can be seen in Figure 114 below.

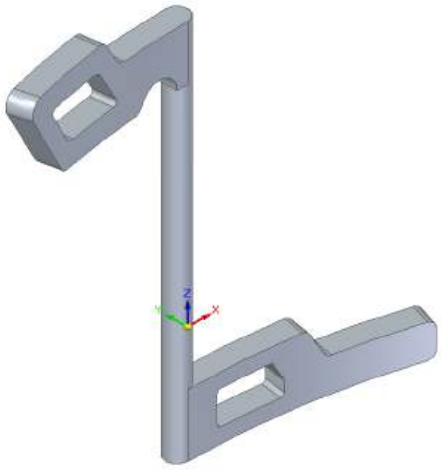


Figure 114: *Cast version of the 180 flag*

Circular full-twist

For the Circular full+twist (CFT) the 180 flag is manufactured with press cut sheet of 1 millimetre thick steel. One end of the flag is held with the other end rotated 360 degrees. The design can be seen in Figure 115 below.



Figure 115: *The 180 Flag, circular full-twist version*

Circular semi-twist

Similar to the CFT the Circular semi-twist is manufactured with 1 millimetre thick steel. The press cut however, has both the top and bottom flag on the same side.The sheet form is then rotated about the flag pole. The design can be seen in figure 116 below.



Figure 116: *The 180 Flag, circular semi-twist version*

U-Shape

The U-Shape proposal reduces a step for the manufacturing process. Both the form and bend of the 180 flag is created a press cut. The 95 millimetre radius of the 180 flag is pressed with a negative and positive die and the flag pole is pressed into a U-shape. The design is shown in Figure 117 below.



Figure 117: *The 180 Flag U-shape version*

Flat press shape

Final and chosen manufacture method for the 180 flag was the Flat press shape. This method is similar to the U-shape. The component is made in one process, pressing the radius while cutting the form of the 180 flag. The flag pole however is left square.

12.7 27 TSK 180 flag

To potentially limit the tool costs for the new design, the 180 flag curvature was increased to fit the TSK 27. This version was then 3D printed and used in a TSK 25. This combination can be seen in Figure 118 below.



Figure 118: *The TSK 25 using flags with the radius of the TSK 27*

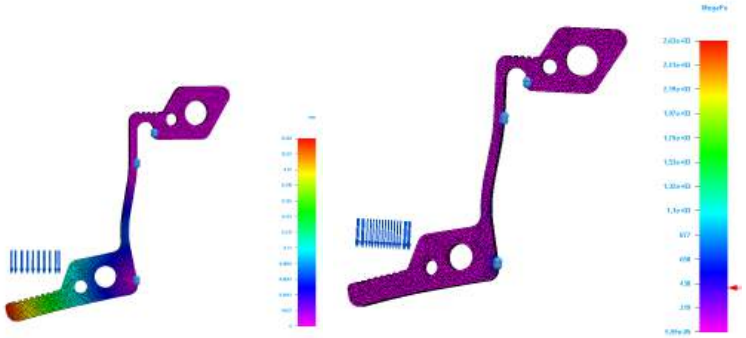
From this 3D print it can be determined to a reasonable level that a 27 sized 180 flag will function within a 25 TSK. This however, will need additional testing, for how the 180 flags are, in die cut metal.

12.8 Strength analysis

To confirm the construction of the new container supports, a strength test was performed in the CAD software Solid Edge. The material was chosen to a default stainless steel with a yield strength of 420 MPa. Supports were placed where the component would be held in the UWS to mimic the real life load scenario as closely as possible. These are represented by the blue dots in Figure 119a, which can be seen below. It is worth noting that the bending of the part seen in Figures 119a - 122b below are exaggerated for visual representation.

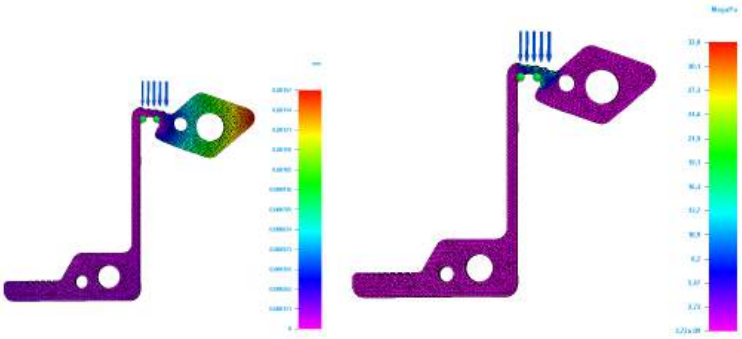
The simulations were done for the pot support and pan support separately to get a good view of how the different parts would perform. Each part was subject to two different load scenarios; one extreme scenario and one representing normal use. The extreme scenario was based on an American standard for camping stoves which proposed a maximum load of 9,1 [27]. However, it was unclear whether this meant per support, or combined load. Subsequently, the load was set to 10 kg per support. The normal use case was based on the largest pot, filled to the brim with water. This resulted in a combined load case of 1,7 kg, or 0,56 kg per support.

The results from the simulations can be seen in Figures 119a - 122b below.



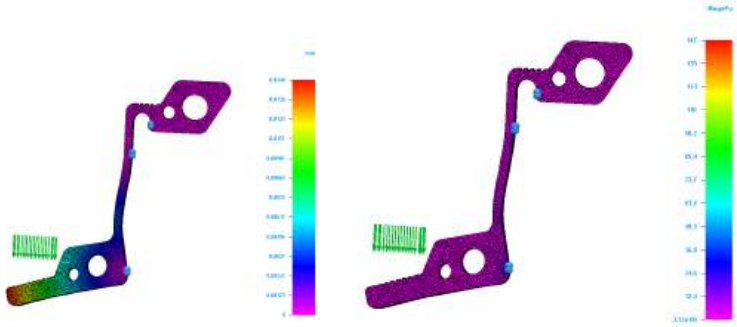
(a) *Deflection (mm) with 10 kg on pot support*

(b) *Stress (MPa) with 10 kg on pot support*



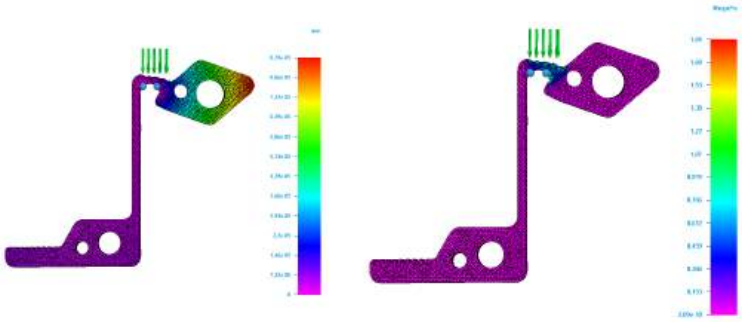
(a) *Deflection (mm) with 10 kg on pan support*

(b) *Stress (MPa) with 10 kg on pan support*



(a) *Deflection (mm) with 1,7 kg on pot support*

(b) *Stress (MPa) with full 1,7 kg on pot support*



(a) *Deflection (mm) with 1,7 kg on pan support*

(b) *Stress (MPa) with 1,7 kg on pan support*

It could be concluded that the construction of the new CS was strong enough to handle the expected load case. The requirements being that no permanent deformation was allowed, and a mamimum deflection of o,5 millimetres. The maximum yield stress was measured to be around 200 MPa in the extreme case, which is far below the maximum allowed strain. During the same simulation, the deflection was measured at 0,264 millimetres, which was well within the specification.

During the expected use case, negligible stress and deformation was experienced for both pot support and pan support.

12.9 Assembly

To provide a complete solution, an assembly plan for the components had to be provided to Trangia. The aim was to develop an assembly method that makes the process as efficient as possible. This would result in saving time during production and in increasing the consistency and quality of the product. Currently, the final assembly is carried out by one person per TSK, which was also the goal of the new solution.

During initial prototyping, using 3D-printed components, it was apparent that it would be challenging to assemble the three CS. The main issue was positioning the CS while adding the ring. The assembly process used during prototyping is shown

in Figure 123 below.



Figure 123: *Assembly of the Flat Ring and 180 Flag in the UWS*

Since there are three CS, which are not attached to any other parts, there was no way to fix them while inserting the ring. The supports have to be held at a certain distance to fit the holders in the ring, otherwise they would not fit. The ring in turn has to be positioned correctly to fit over the CS for correct function. This resulted in the idea of creating an assembly jig that would hold the CS in place, while providing a guide for inserting the ring in the UWS. The result is shown in Figure 124 below.



Figure 124: *Proposed design for assembling the CS and flat ring*

The assembly jig (light blue ring in Figure 124) shown above is used to hold the CS

in the correct position. The three CS will be placed in the jig, which will then be put on top of the UWS. The flat ring will then be compressed and put within the jig. There is a guide for aligning the support holders on the ring with the CS. A plunger is then used to push the ring into the UWS, and it will snap into the correct position.

12.10 Cost analysis

A rough cost analysis was performed to compare the new solution to the existing one. This was done using information provided from Trangia regarding current manufacturing. The constants used for the cost analysis are sensitive to Trangia. Due to privacy reasons, the full extent of the information cannot be disclosed in this report. However, the new solution was estimated to be both cheaper and faster to produce.

12.11 Complete design- Weight

To meet the design requirements the 180 flag concept was required to be a maximum of 50 grams heavier compared to the original design. Table 10 compares the weight difference between the two solutions.

Table 10: Comparison between the original and new support components.

Component	**Old (g)**	**New (g)**	**Difference (g)**
CS	6.7	9.2	2.5
Ring 25	59.7	26.7	-33.0
Combined	79.8	54.3	-25.5

The new CS is 2.5 grams heavier than the existing CS. This was expected as the hooks on the TSK are minimal and light. The new ring design however is 33 grams lighter, due to its manufacture from thin sheet metal steel. Even though three 180 flags are 7.5 grams heavier than the original hooks the complete design is 25.5 grams lighter than the existing product.

13 Final design

Within the following section the finalised design of the 180 flag concept is presented.

13.1 The concept

The design uses the same principles as the existing TSK. The number of parts is identical with three CS and a ring securing them. The separate components can be seen in the exploded view in Figure 125 below.



Figure 125: *Exploded view of the new design*

13.2 Material

Both components will be made of stainless steel to withstand the rough conditions. The 180 Flag will use the same material currently used for the CS, and the ring will be made from spring steel. The materials are presented in Table 11 below.

Table 11: Materials for the 180 Flag and the Flat Ring

Component	Material
180 Flag	Stainless Steel 1.4301
Flat ring	Stainless Steel 1.4310

13.3 Variation between the 25 and 27

The design is intended for use in both the 25 and 27 TSK's. The 25 and 27 can use the same 180 flags but different rings. The length of the ring is reduced to fit the smaller circumference for the 27 TSK. The new design for the 25 and 27 TSK can be seen in Figure 126 below.



Figure 126: *The new design of the 25 and 27 TSK*

13.4 5 Stages of Use

For this design the number of stages for use are reduced, making the user interaction simpler and easier. When creating a meal that requires both the Pan and Pot, the user doesn't have to change the CS, fiddling with a fork on the hot metal. In Figure 127 below the 5 stages of use are presented.



Figure 127: *The five stages of use*

During the first stage the pots are housed within the UWS. They can then be removed, lifted out. The hooks are then lifted, rotated 90 degrees counter-clockwise and placed on the UWS ridge. Either the pot or pan is then placed, ready for use.

13.5 The 180 flag component

The 180 flag is comprised of a flag pole with two non identical flags 180 degrees in rotation from each-other. The part is made from hardened steel. The final design can be seen in Figure 128 below.

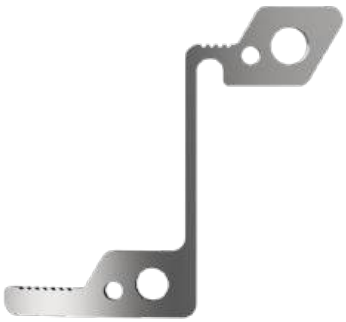


Figure 128: *Close up of the 180 flag*

Technical drawings of the 180 flag can be found in Appendix B .

13.6 The Flat ring component

The Flat ring design uses three pressed archs to house the 180 flag component. The component is made from spring steel. The final design can be seen in Figure 129 below.



Figure 129: *Close up of the Flat ring*

Technical drawings of the Flat ring can be seen in Appendix B.

13.7 The new UWS

The new UWS requires a lowered groove distance and due to the form of the flat ring a square form. The groove internal form is 5,6 millimetres to fit the flat ring. The groove from centre is 18,9 millimetres from the peak of the upper ridge. A close up of the new UWS can be seen in Figure 130.



Figure 130: *Close up of the new UWS groove*

A technical drawing of the new UWS can be found in Appendix B.

13.8 The complete design

Figure 131 below presents the complete design in the anniversary celebration 'Cloudberry' orange.



Figure 131: *The new design of the TSK in cloudberry*

14 Testing final design

This chapter presents the final testing of the design to evaluate its functionality and efficiency. This was done through experiments, real life testing and a short user

study.

14.1 The prototype

For this final design test, the prototype was created using materials as close to feasible to create an accurate representation of the final design. The prototype shown in Figure 132 below uses the correct spring steel. The 180 flags are however, made from softer steel and are rougher than expected. The ring had to be formed by hand and is therefore not fully accurate. Additionally, the ring is positioned in the original UWS, resulting in the flat ring sitting in a round groove. These discrepancies limit the functionality and hinder the user study, but are negligible for the experiment.



Figure 132: *The UWS with the new flat ring and 180 flags*

14.2 Experiment

Since the pan sat slightly lower with the new design, it was deemed appropriate to test how this affected heat transfer when cooking. Consequently, two experiments were carried out, one using the spirit burner and one using the gas burner.

The gas burner experiment followed the method presented in 8.6. However, the current CS and the 180 Flag were used for the different pan heights. The control knob was also turned to 4/3 open. The results of the experiment are presented in Figure 133 below.

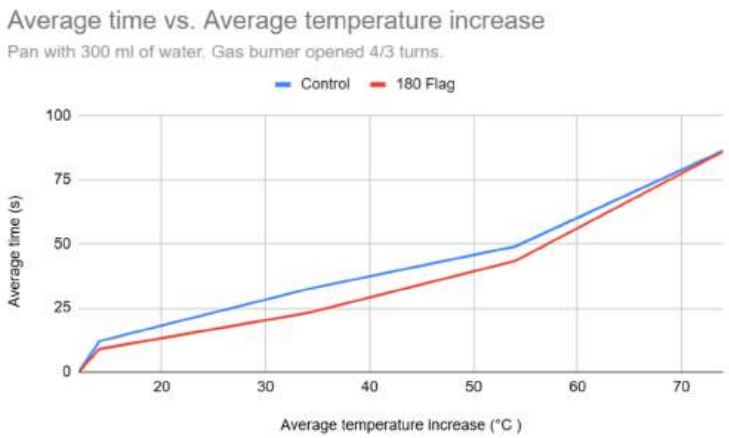


Figure 133: *Results from experiment heating 300 ml of water with gas burner using the 180 Flag and current CS as supports*

As can be seen in the graph above, there was no considerable difference between the current CS and the 180 Flag. However, this was not unexpected as the position had been lowered by only a few millimetres. The main takeaway from this experiment was that the heat exchange had not been affected negatively when using the 180 Flag.

The spirit burner experiment followed the method presented in 8.7, with the 180 Flag and current CS providing the two pan heights. The results of the experiment are presented in Figure 134 below.

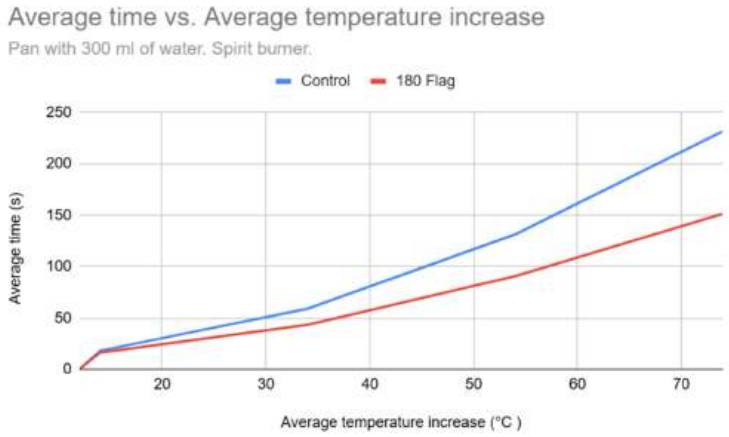


Figure 134: *Results from experiment heating 300 ml of water with spirit burner using the 180 Flag and current CS as supports*

The results of the spirit burner test indicated that using the 180 Flag decreased the heating time of the water. However, the results were fairly inconsistent with a lot of variation in time for the control. As with the gas test, it indicated that the use of the 180 Flag did not negatively affect performance.

14.3 Real life testing

To properly evaluate the design the new version of the TSK was tested in the woods and used during a feasible activity where a TSK would be useful.

The TSK was tested during a short day trip for bouldering. This involved testing the product from travelling to the bouldering spot, preparing, cooking food and making tea. During the walk to the boulders the TSK didn't appear to rattle much. The 180 flags are expected to be far less mobile when packed compared to the original CS. The set up during cooking can be seen in the Figure 135 below.



Figure 135: *Cooking sausages by some boulders*

The new design was proven to function well during cooking. In Figure 14.3 below, the 180 flag holding the pan can be seen.



(a) *Placing of the pan on the 180 flags*



(b) *Pan placed and sausages cooking*

It can be noted that there could be an advantage in increasing the length of the 180 flags top angled step. The pan did however feel secure and not difficult to place on the 180 flags.

The kettle functioned very well with the new CS. Its use can be seen in Figure 14.3 below.



(a) *Placing of the kettle on the 180 flags*

(b) *Kettle placed and boiling*

There were no issues in using the kettle with the new TSK.

14.4 User study

Though the length of the project did not provide enough time to do a full user study, a preliminary study was made. This involved testing the new TSK on four different people, all with experience using Storm kitchens and outdoor cooking.

Listed below are key quotes from these four people on the new TSK.

- J- "Oh it goes that way" (Rotating the 180 flag)
- D- "Nice and Sturdy" (holding the UWS)
- S- "Quite stiff" (lifting the 180 flag)
- O- "Are they supposed to go down like that" (height for the 180 flags)

Though the new design is only a prototype the preliminary user study has assisted in understanding how the design might be enhance. For all of the users they found the design robust and required force to use. With a more precise manufacture for a future prototype any stiffness could be nullified.

As designers, this preliminary study has been very useful. Users interact with the product differently than expected. The 4th user's test, 'O' is shown in Figure 14.4 below. This user helped us realise that, as they didn't automatically assume that the 180 flags rested on the UWS ridge.



(a) *O placing the pan on the 180 flags*



(b) *O comparing ease of removing pots between the two versions*

It was also determined that removing the pots from the UWS when packed was easier with the new version.

14.5 Requirement specification evaluation

The 180 flag final design was evaluated against the Requirement specification with the aim to assess the feasibility and success of the design. The evaluation can be seen in Table 12 below.

Table 12: Requirement specification-evaluation

No.	Statement	Required	Wish	Achieved
1	The improved solution should be compatible with all previous versions of the storm kitchen.	x		x
2	The weight of the new solution should not be higher than the current one.		x	x
3	The weight should not increase by more than 50 grams.	x		x
4	The improved solution should not be more difficult to use than the current one.	x		x
5	The manufacturing cost should not be increased.	x		x
6	The improved solution should make use of current manufacturing techniques available to Trangia.		x	x
7	An investment in new machines should not exceed the given budget.	x		x
8	The cooking efficiency should be improved.		x	-
9	The new solution should not contain any loose parts.	x		x
10	Reducing production time.	x		x
11	Improving manufacturing consistency (tolerances).	x		x
12	The container supports must sustain 9,1 kg.	x		x
13	The supports should work with containers with smaller diameter.		x	x
14	Prevent the rattling inside the storm kitchen when packed.		x	
15	Make the supports compatible with new cooking accessories.		x	
16	The improved solution should be more user friendly than the current one.		x	x
17	The coffee pot must fit inside the kitchen when packed.	x		x

Overall, based off the Requirement specification, the 180 flag concept can be seen as a success. All of the statements that were considered "Required" have been met.

For No.14 and No.15 the statements were wishes. For both of these statements the design can be slightly redesigned to be achieved. The rattling of the pots and pans has not been removed but an attachment or modification could likely fix this. For the cooking accessories it is difficult to confirm as technical drawings are unavailable to confirm the compatibility.

15 Discussion

The discussion is presented below as several sections. These selections are areas where the design requires further development, plus reflections on the project.

15.1 Tooling/manufacturing

When creating prototypes for both 3D printing and Waterjet cutting a key limitation has been the tolerance issues with these processes. For future development the design needs to be tested with manufacturing methods that have finer tolerances.

It is unclear whether the flat rings spring steel can be manufactured with press cutting. This unfortunately cannot be tested easily. Additionally, using die cutting a prototype 180 flag can be accurately tested to fit correctly within the UWS and against a die cut Flat ring.

15.2 Prototyping

Though the design has been tested with almost the correct materials, its hard to create a prototype that perfectly reflects the realised concept. The key limitations for this has been the access to manufacturing methods and costs that come with such methods.

Once a prototype where the components have been press cut, tumbled and treated, complete testing can be undergone. It will be important to test whether the Steel 180 flags will scratch the aluminium lining of the UWS. Another concern is how the 1880 flags may rattle when packed.

The design also should be tested long term. As Trangia is proudly producing products that last a "lifetime or three" [1] ,its important to test this concept over a longer period than is feasible for this project. This also relates to testing the sturdiness of the 180 flag concept. The 180 flag is unlikely to degrade over time, but must be further tested for how sturdy the joints between parts will be over time.

15.3 User Study

Though the product has been tested for other users, the number of users has been very limited. It would be advisable to test the concept on a far larger number of

users. This could find additional potential issues and improvements.

One possible consideration for concept testing was a secondary survey comparing the original design to the new design. This however would breach the NDA. A controlled survey run by Trangia could lead to a quantitative user opinion and be a useful tool for further improvements. This survey could test how experienced users would adapt to a new design as well as understanding the learning curve for new users.

15.4 Personal reflections

The design has met the design brief and also covered the majority of the requirements listed in the specification. This project can be a considered a success, and a development of a functional and appropriate solution for Trangia's requirements.

It can be reflected that the methods followed throughout the project have worked well. There are however, areas that could be seen as lacking. During the research phase, observations and users studies were remiss. This was deemed allowable as the brief focused primarily on manufacture and not user interaction. As the project developed, this brief was drawn closer towards user interaction. So upon reflection, observations and user studies could have been a benefit.

16 Conclusion

The final solution is a complete redesign of the Trangia CS. It uses a new principle where the load of the cookware is transferred to the UWS, removing strain from other components. The compatibility with previous versions of the TSK has been ensured, but with an improved user experience.

The fundamental requirements for the component have been fulfilled with the new solution. The basic functions remain unchanged, and the manufacturing can be performed using existing machines in the Trangia factory. The design aims to communicate Trangia's brand through familiar design elements and details.

This report describes an exploration of alternative solutions for CS. The final result should be seen as a proof of concept, and not a fully developed solution. However, it provides the basis for final refinement, and implementation, of the concept.

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Appendix

A User study Survey

Page 1 **??**

Below is a list of the websites where the survey was posted with varying degrees of success for answers:

Facebook groups:

1. Camping i Sverige - tips på campingplatser och annat som rör campinglivet
2. Trangia lovers
3. Stormkökets vänner
4. Survival Bushcraft
5. Wild Camping UK
6. UK Wild Camping: Bushcraft, Hiking, and Fire cooking association
7. Camping and Bushcraft
8. Outdoor cooking and Rocket stoves

Reddit:

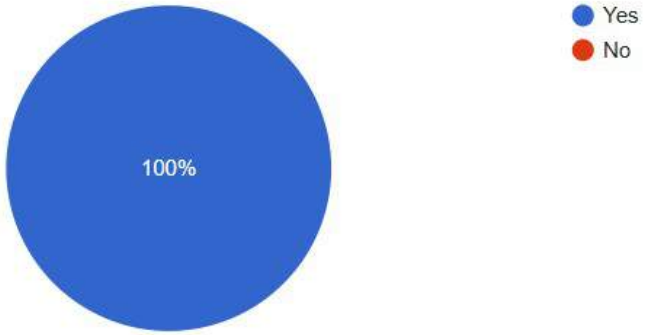
1. r/trangia
2. r/outdoorcooking
3. r/DofE
4. r/hikinggear

Utsidan.se

Presented below is a screenshot of the google form of the survey.

Have you used a Trangia Storm kitchen?

140 responses



Have you ever had issues with the containers' supports on the Trangia Storm kitchen (i.e. longevity durability)?

140 responses

No although the hindges can be a little loose. Doesn't affect use.
No, never
No issues so far
Some deformation on the windscreen from holding heavier pots. No issues affecting use though.
Not really, apart from the odd bend through misuse
No,have had Trangia since 1960 and have 4 new ones newer any problem. Have Optimus Jägarkök and Esbit about the same, pots are cone shaped and resting on the rim so no holders,
No, they work perfectly
Too stiff too fold up and down easily
No, - they work well even after many years

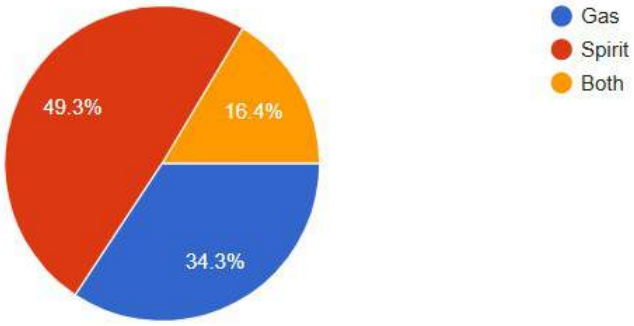
Do you feel other brands have a better solution for holding different sized containers?

140 responses

Maybe, although I'm not aware of other brands with spirit burners enclose the pot or container with a windscreen
Not really. Trangias are legendary for stability. I have however recently bought a Trangia griddle so I can use a mokapot coffee maker on it. A biycle brake disc works just as well, but doesn't pack so neatly.
Yes, but they're usually not sold together as a full kit.
Not really - tradeoff between range of sizes accepted and stability.
Se above question but haven't find one yet with better than Trangia!
Not to my knowledge
Not for comparable stoves.
I think the the Trangia pot stand is too heavy

Do you use a gas or a spirit burner?

140 responses



Gas burner focused questions

Why do you prefer using a gas burner?

7 responses

I am not sure of the fuel quality that I find in my country. I don't want toxic stuff in my food so I use gas to be sure

Find it easier to transport gas cylinders (though can be more difficult if travelling internationally) and the flame intensity is better with gas

Faster, cleaner, lighter

Ease of temperature adjustment. Only burner allowed in NSW Australia during a Total Fire Ban. Less smell than metho.

Don't like carrying spirit from environmental / safety perspective. Used to use gel fuel but more difficult to source and so swapped to gas.

Less Mess, faster boil tine

Safety and control. We are not allowed to use spirit burners after some nasty accidents (operator misuse). I have used meths in the past personally and much prefer the ease of use with gas.

Spirit burner focused questions

Why do you use a spirit burner?

92 svar

You can see how much fuel you have.

I use both, but spirit because it is easy to see how much is left, feels safer, with gas I am always worried to ran out

Yeah

Because it's silent and the fuel is available in most regions

It's so silent! And fuel is easy available.

Eco friendly (but do also use a gas burner occasionally, but couldn't say both on your questionnaire.

Works

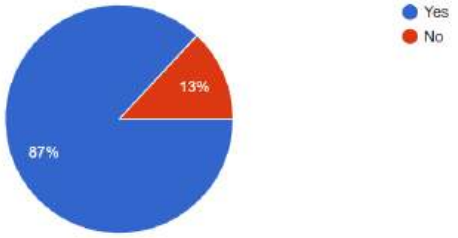
In the bush Less to pack in and out, use also Gas in our kitchen when there is an electric outage.

It is useful, and I love the smell of the burning spirit. I do have a gasburner, and I use it some times.

Do you use the Trangia simmer ring to control the flame?

Kopiera diagram

92 svar



Why do or don't you use the simmer feature?

92 svar

To control the flame
To control the heat output.
It helps the heat level
To control the flame and to save fuel
to simmer my food and prevent overcooked food
Although slightly basic, after 40 years, I've got the hang of it. I also have a Trangia mountain marathon stove, which although using the same burner and simmer ring, is best used for water boiling.
Control intensity of flame
I said yes that I do so why do I have to fill this in!!!!
I do use it!

Do you feel other brands have a more user friendly solution for simmer cooking?

92 svar

Yeah

Not for alcohol stoves

not really, as including a handle for the simmer ring prevents the usage of a windscreen , thus often need more power to get the food to a boil.

I also use a homemade catering bean can woodstove, but it has no simmer. My Coleman Peak 1 stove does simmer but the temperature is too low to heat the generator tube effectively.

NO.

I don t know

There is a Chinese copy that has a fold away wire clip to adjust the simmer ring that is much more user friendly

Gas stoves are more user friendly in that regard.

Do you have any other overall thoughts for improvement?

80 svar

No

no

A bigger kettle

More colours

I'm very happy with the 27. No improvement suggestions.

I'd like a slightly bigger kettle for my Trangia 27 but one that would still fit in the nest. The original kettle doesn't quite fill two mugs.
This is a classic stove largely made exactly as it was, and has a great following with young people due to it's stability. My daughter first took mine camping a few years ago. She uses the gas burner, but wouldn't dare take the Coleman flamethrower.
I suspect that the Scouts and D of E campers who have used group kit for decades, often go on to buy their own.
I regularly wild camp with two friends, also 70+, and although we'll have a fire if we can, if we need a brew,

B Design

The following section shows the technical drawings of the components.

Technical drawings

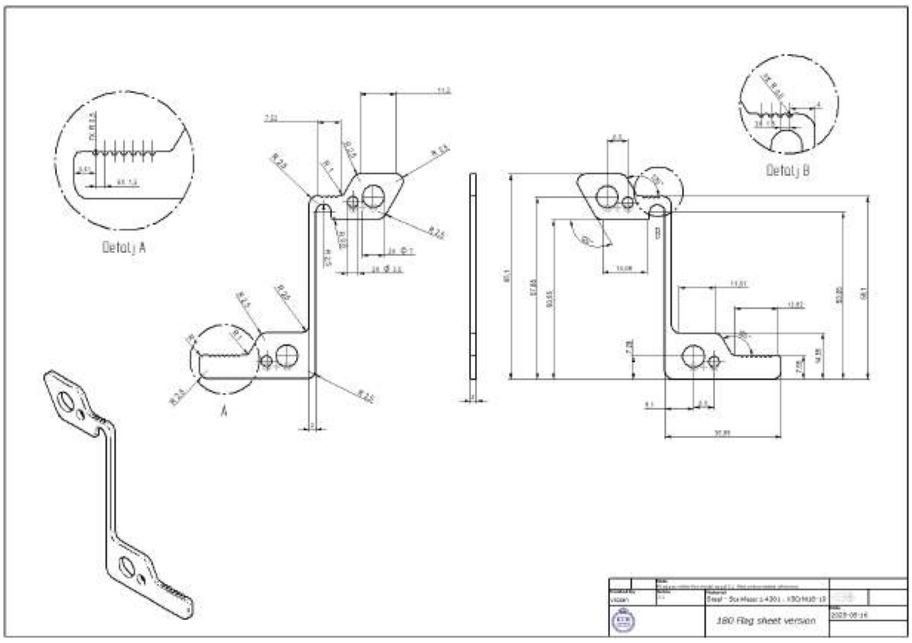


Figure 1: *Technical drawing of the 180 Flag as a flat sheet*

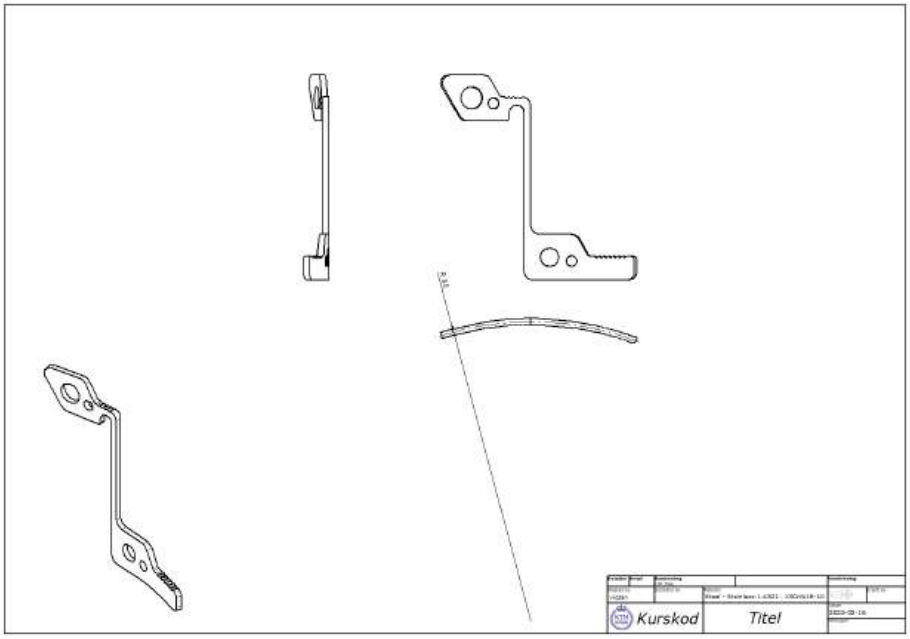


Figure 2: *Technical drawing of the 180 Flag after bending*

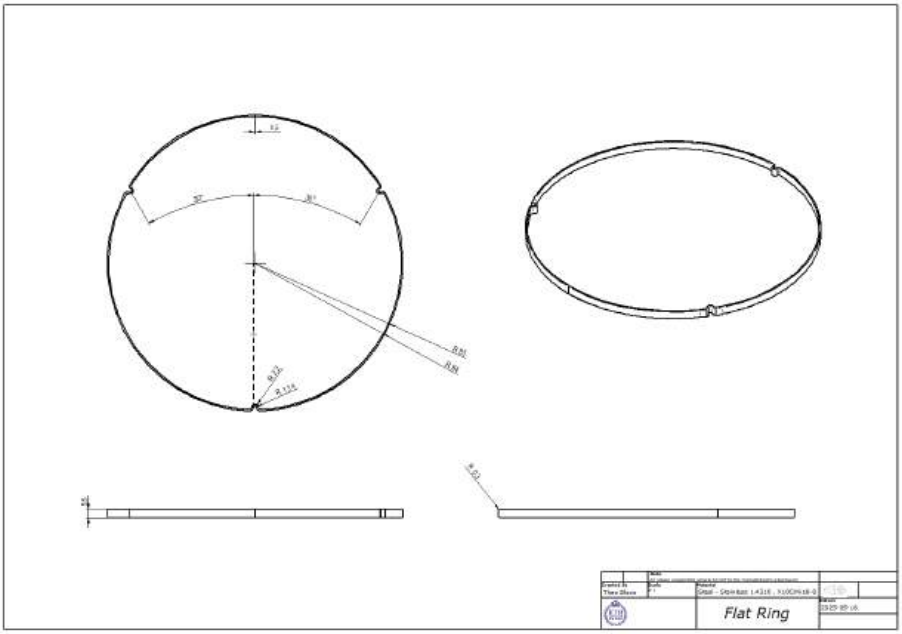


Figure 3: *Technical drawing of the Flat Ring*

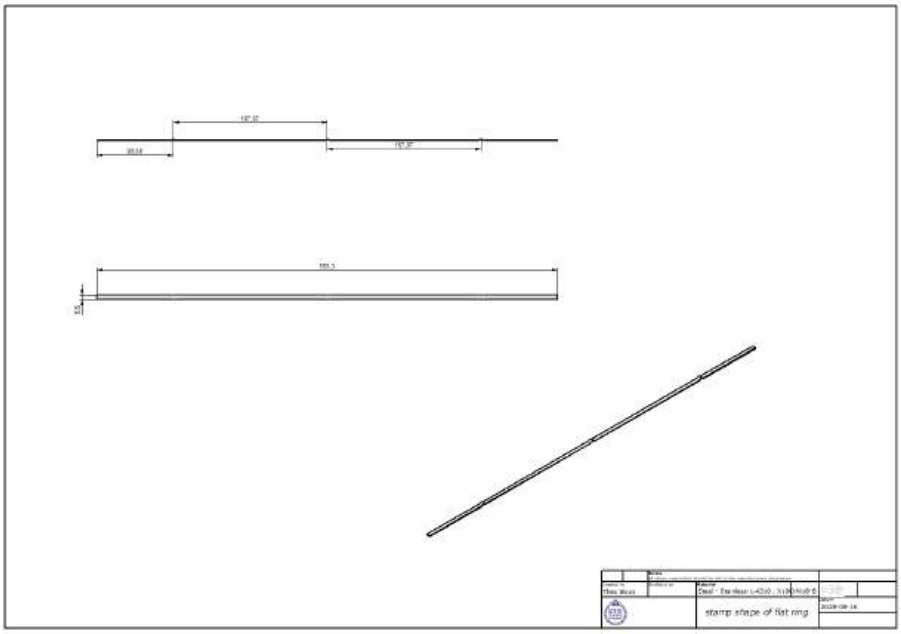


Figure 4: *Technical drawing of the sheet cut Flat Ring*

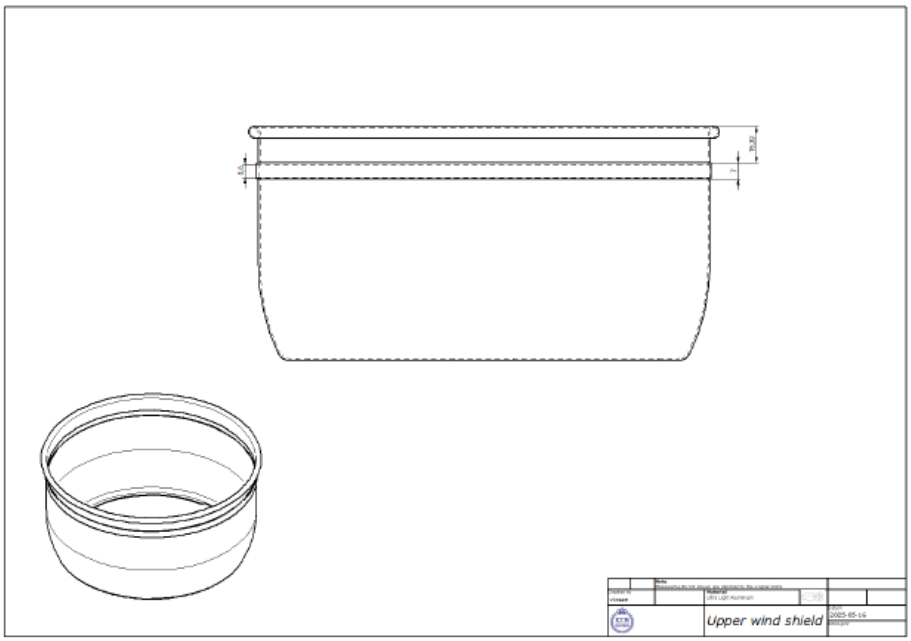


Figure 5: *Technical drawing of the updated UWS*

C Experiments

The following section presents the full data from the conducted experiments.

| Height | Temp 1 | Temp 2 | Temp 3 | Temp 4 | Temp 5 | Temp 6 | Temp 7 | Temp 8 | Temp 9 | Temp 10 | Temp 11 | Temp 12 | Temp 13 | Temp

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